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Inoue et al.

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(54) **FINE WATER DISCHARGING ELEMENT,
FINE WATER DISCHARGE DEVICE AND
METHOD FOR MANUFACTURING FINE
WATER DISCHARGING ELEMENT**

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Primary Examiner — Arthur O. Hall

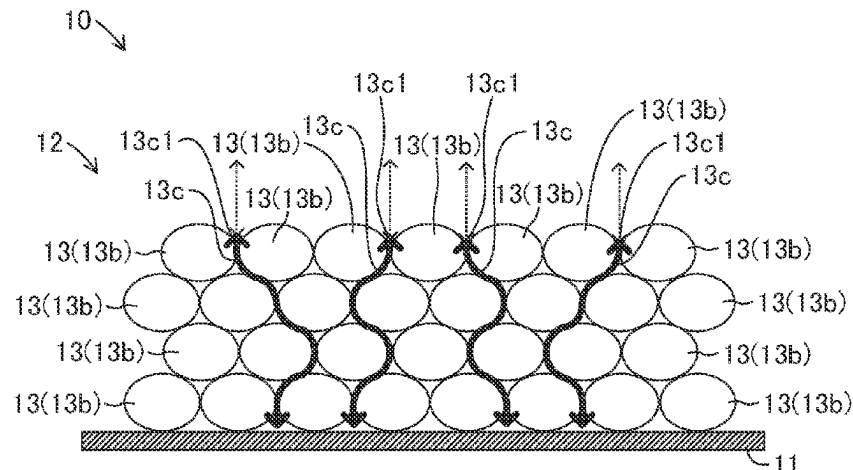
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(57) **ABSTRACT**

The fine water discharging element capable of transitioning
between an adsorption state where water contained in a fluid
to be treated is adsorbed and a discharge state where the
adsorbed water is discharged to the fluid to be treated, the
fine water discharging element comprising a base material
portion, a plurality of particles and a nanochannel formed
between the shell portions by laminating the plurality of
particles on an outer surface of the base material portion in

(Continued)



a rich viscoelasticity film shape. The fine water discharging element is transitioned between the adsorption state and the discharged state by changing the temperature of the water in the nanochannel by controlling an electrifying of at least one of the base material portion and the plurality of particles.

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11 Claims, 11 Drawing Sheets

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G02B 1/14 (2015.01)

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CPC **F24F 6/025** (2013.01); **G02B 1/14**
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CPC G06F 3/041; G06F 2203/04102; A61M
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USPC 345/173

See application file for complete search history.

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Fig.1

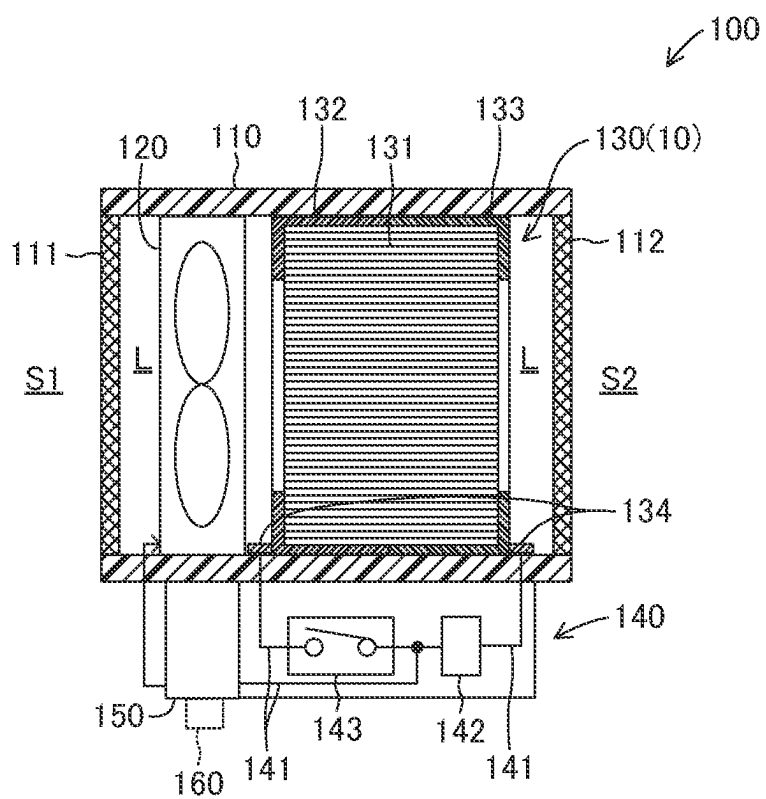


Fig.2

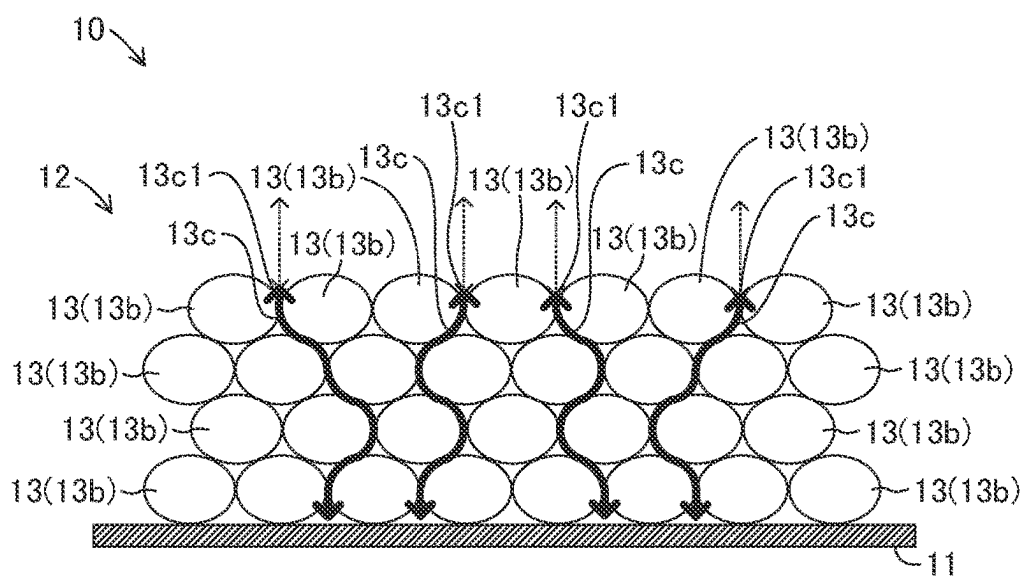


Fig.3

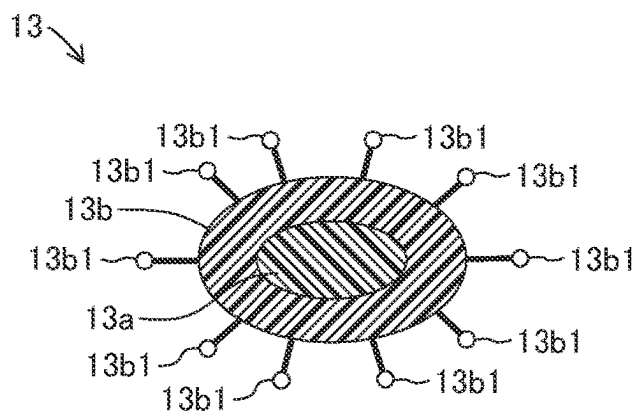


Fig.4

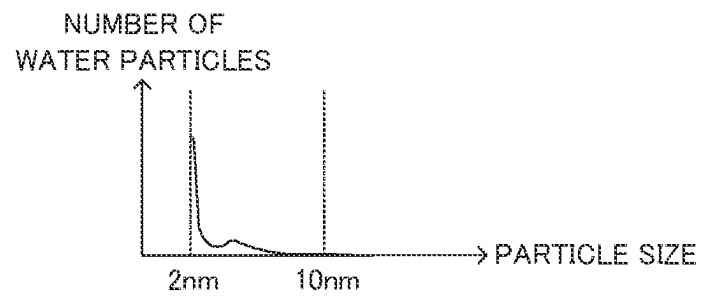


Fig.5

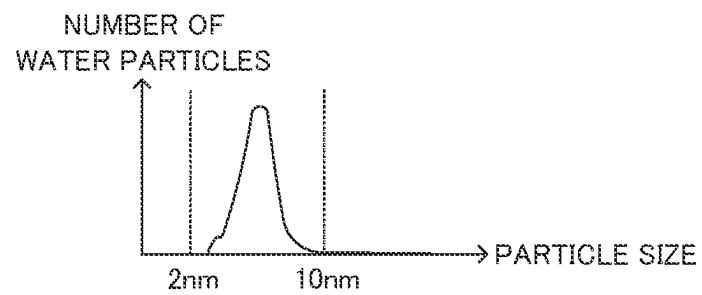


Fig.6

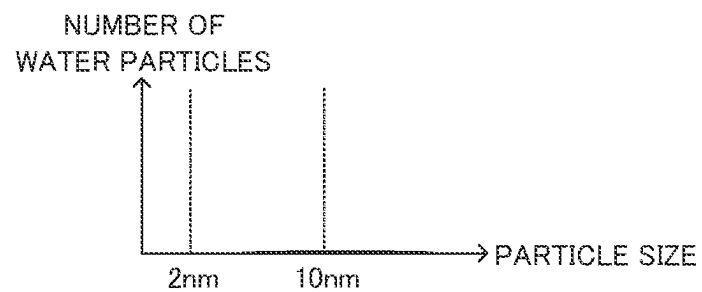


Fig.7

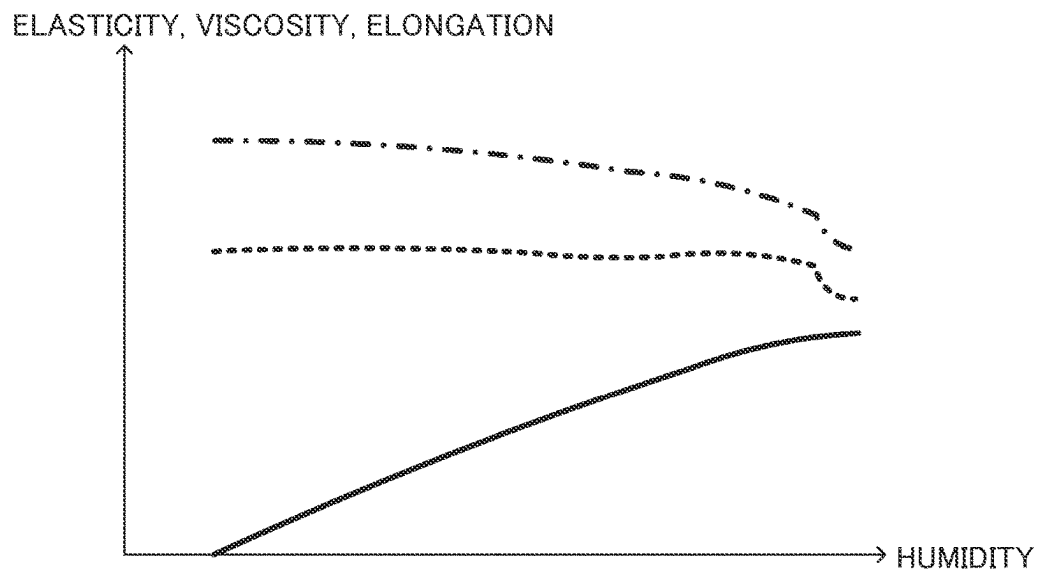


Fig.8

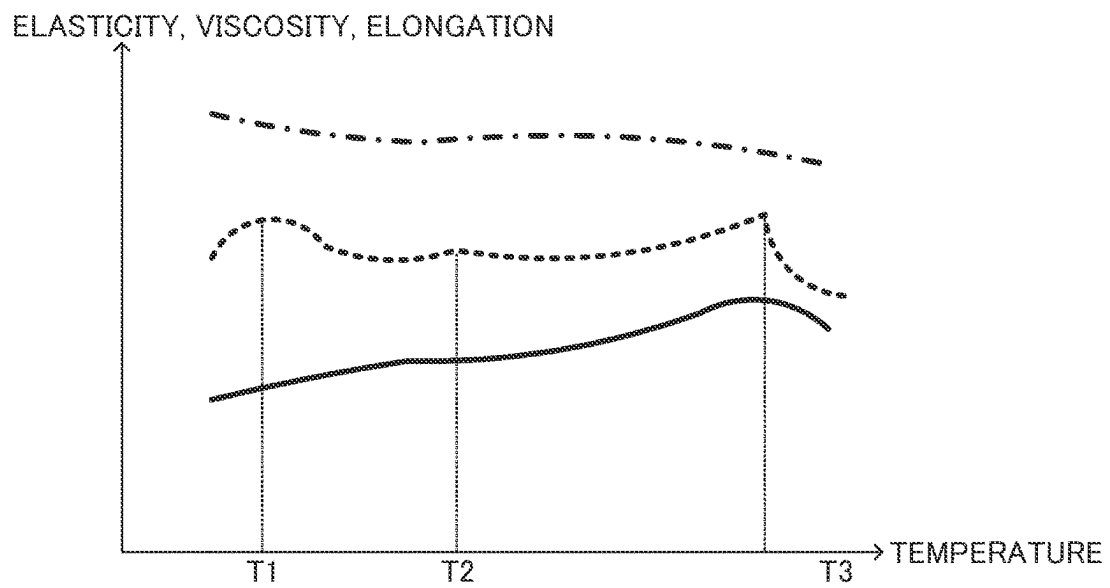


Fig.9

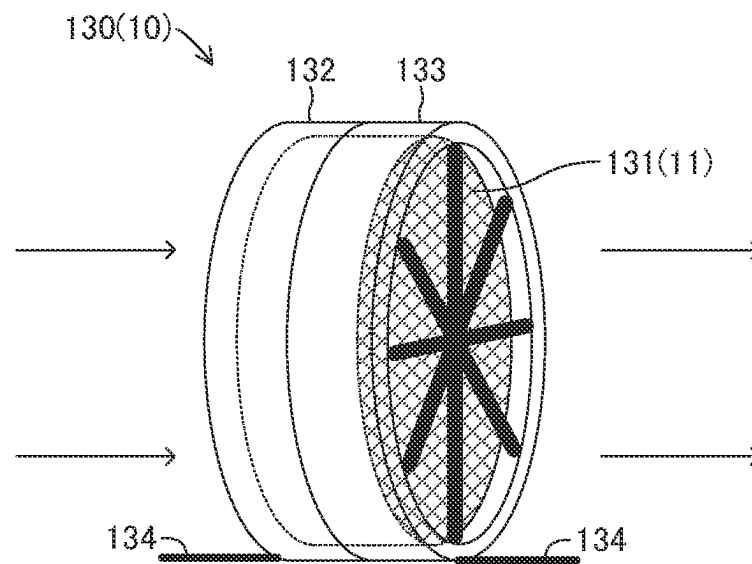


Fig.10

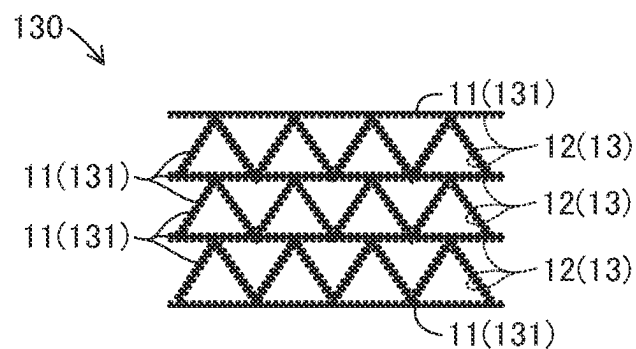


Fig.11

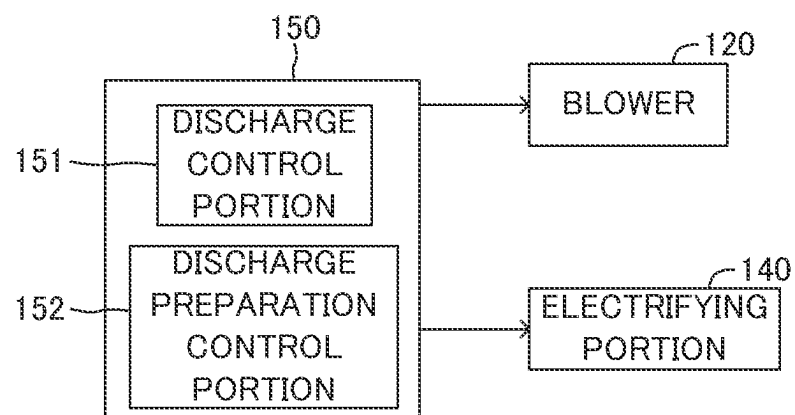


Fig.12

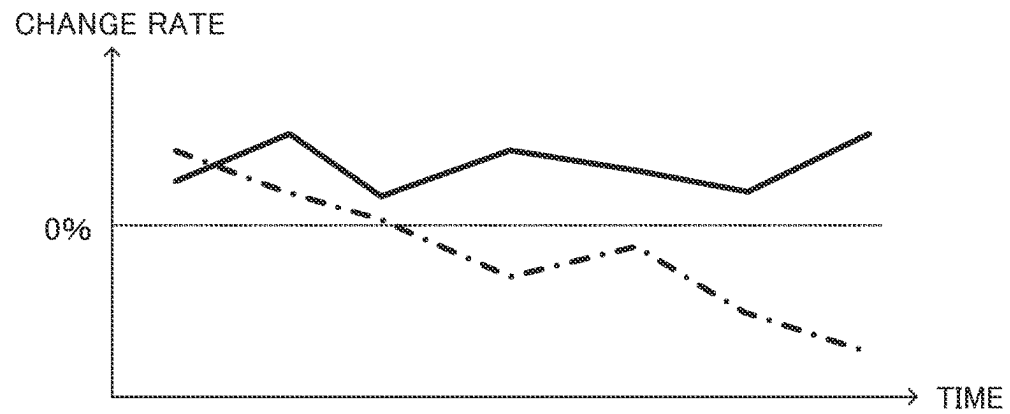


Fig.13

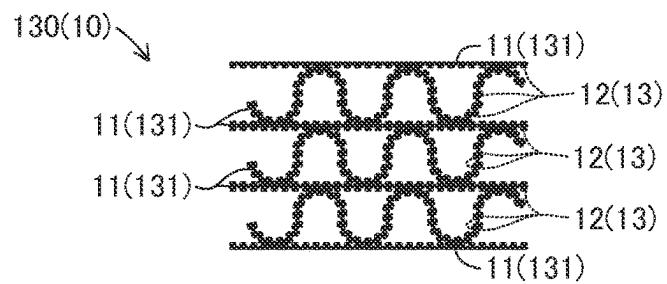


Fig.14

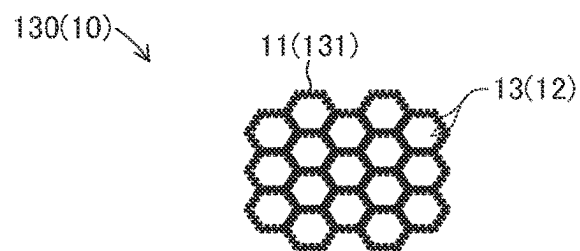


Fig.15

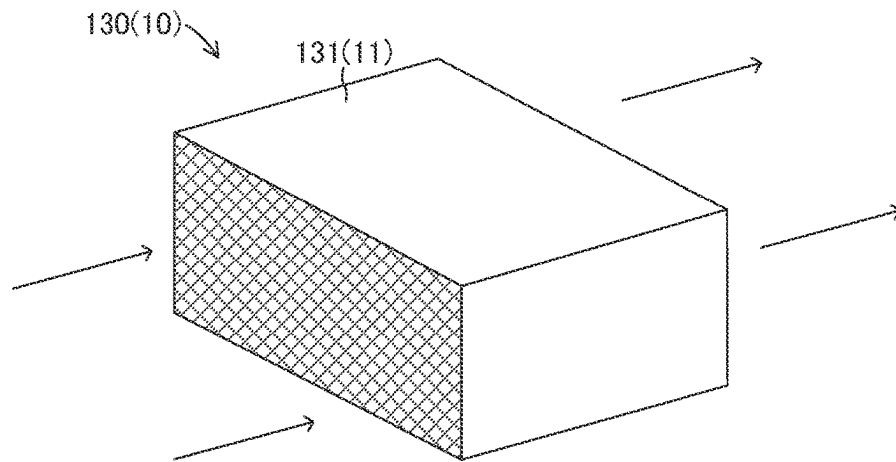


Fig.16

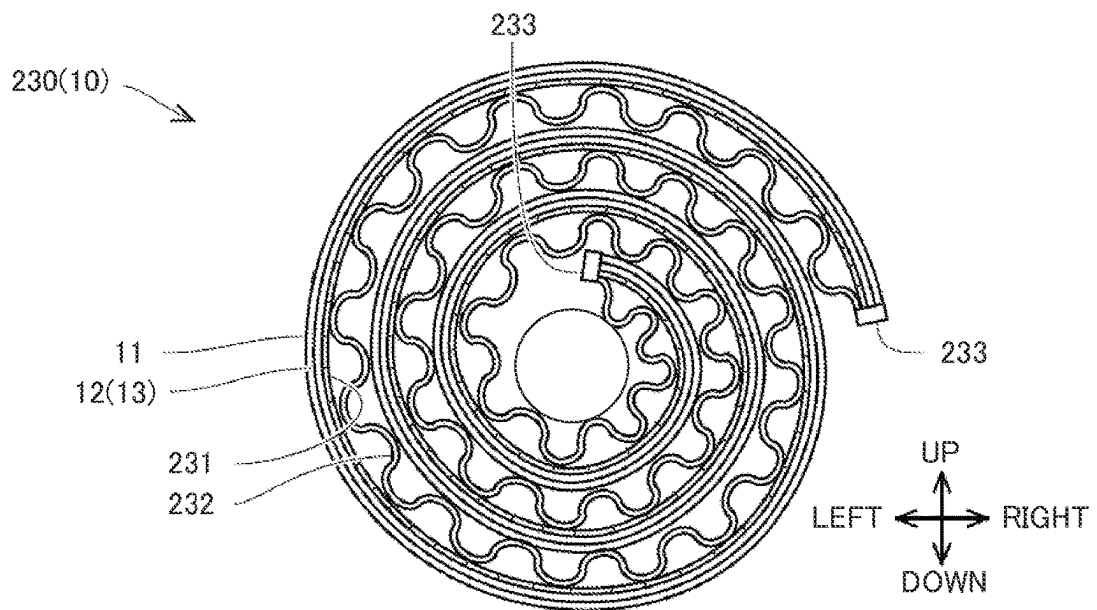


Fig.17

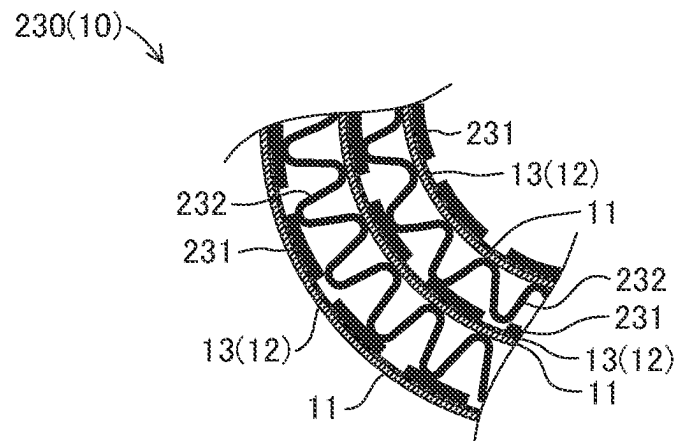


Fig.18

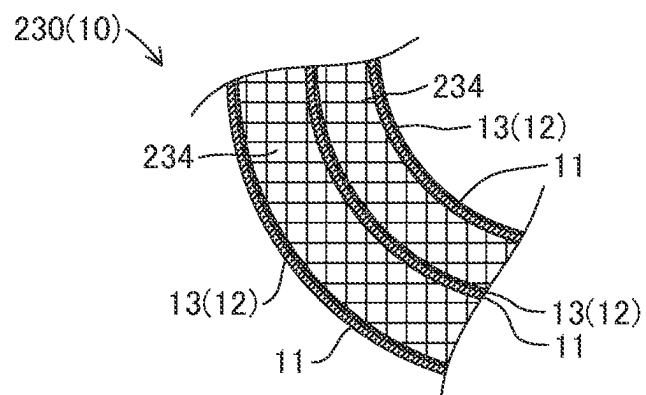


Fig.19

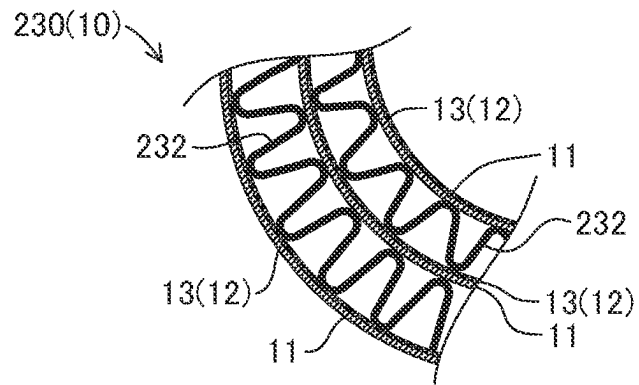


Fig.20

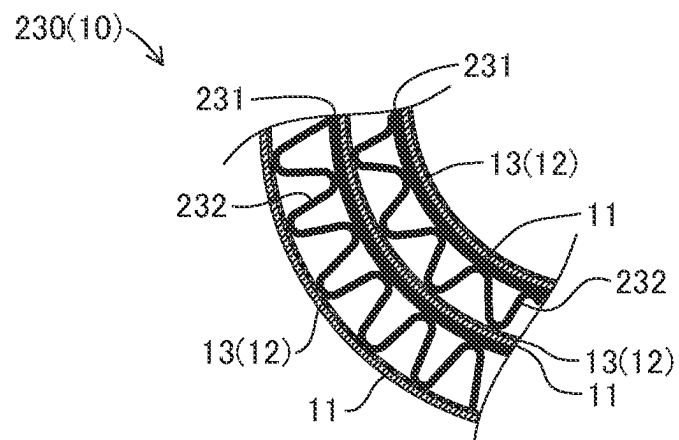


Fig.21

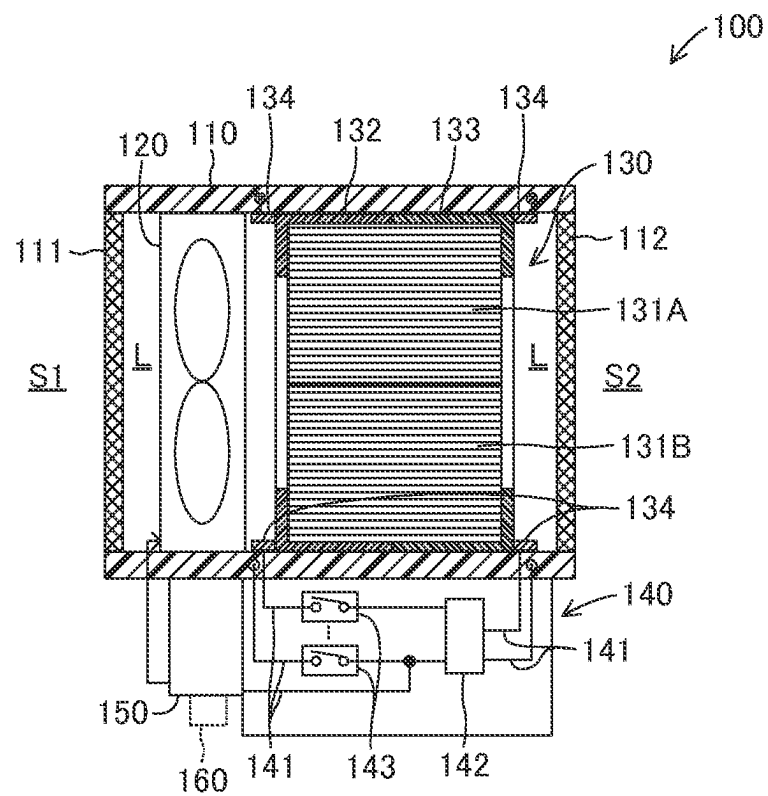


Fig.22

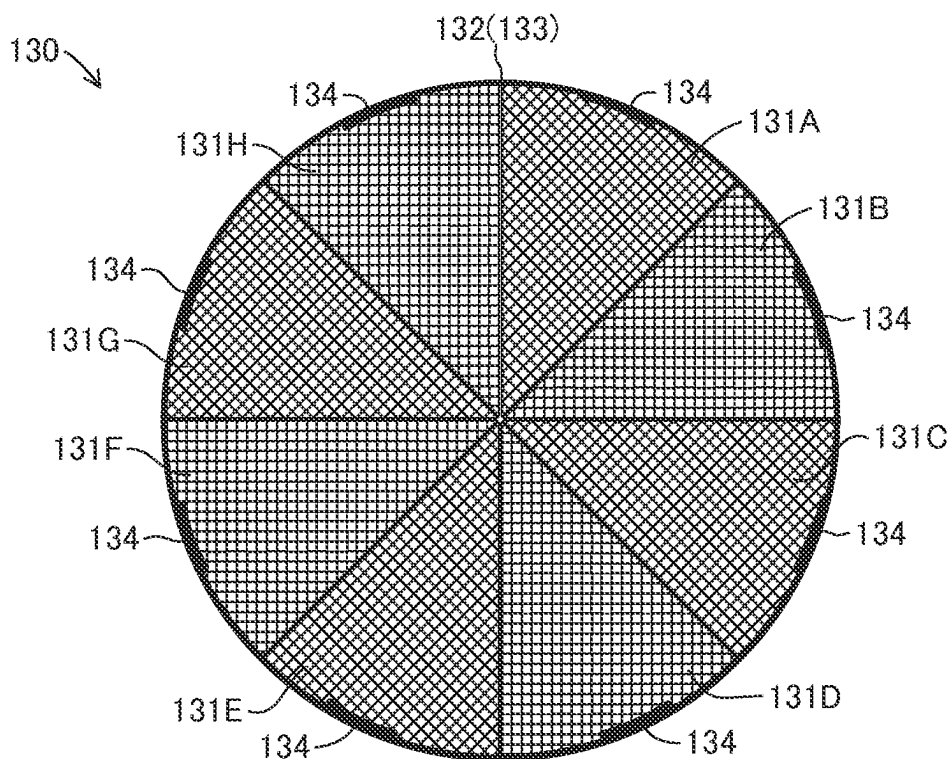
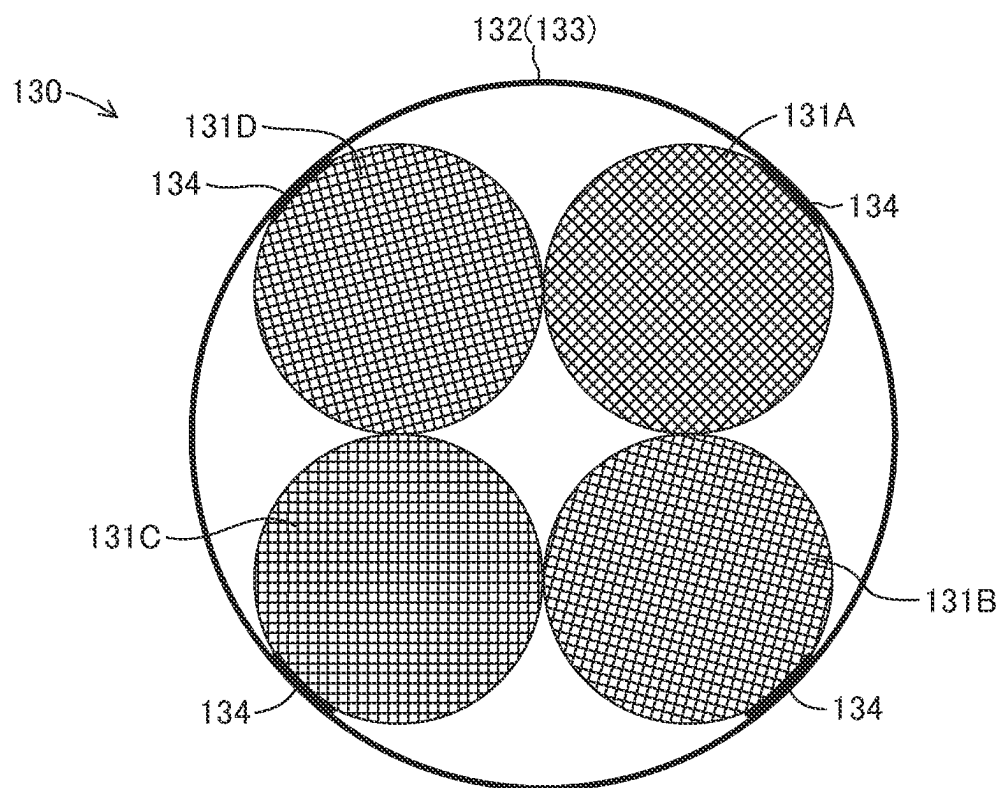


Fig.23



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**FINE WATER DISCHARGING ELEMENT,
FINE WATER DISCHARGE DEVICE AND
METHOD FOR MANUFACTURING FINE
WATER DISCHARGING ELEMENT**

CROSS-REFERENCE TO RELATED
APPLICATIONS

This application is a continuation of U.S. application Ser. No. 17/269,387 filed Feb. 18, 2021, the entire contents of which is incorporated herein by reference. U.S. application Ser. No. 17/269,387 is a 371 of International Application No. PCT/JP2019/007674 filed Feb. 27, 2019, and claims the benefit of priority from prior Japanese Application No. 2018-172166 filed Sep. 14, 2018.

TECHNICAL FIELD

The present invention relates to a fine water discharging element, a fine water discharge device, and a method for manufacturing the fine water discharging element.

BACKGROUND ART

For example, in the related art, an electrostatic atomizer disclosed in PTL 1 below is known. The electrostatic atomizer of the related art includes a water transport unit which transports water by a capillary phenomenon, a heat exchange unit which supplies condensed water generated by cooling air on a heat absorbing surface to the water transport unit, an application electrode which applies a voltage to the water transported by the water transport unit, a counter electrode which is located to face the water transport unit, and a high voltage application unit which applies a high voltage between the application electrode and the counter electrode. Then, in the electrostatic atomizer of the related art, water retained at a distal end portion of the water transport unit is atomized toward the counter electrode by the high voltage applied by the high voltage application unit, and thus, nano-ion mist having a nano-sized and strong electric charge is generated.

Further, in the related art, for example, a negative ion and nano-mist generator disclosed in PTL 2 below is also known. The nano-mist generator of the related art includes a treatment chamber through which outside air passes by a feeder, a water storage section which is provided in a lower portion of the treatment chamber, a mortar-shaped rotation member of which a lower portion is submerged in the water storage section and a diameter increases upward, and a cylindrical porous body which is located on an outer periphery of the rotation member and rotates together with the rotation member and through which water and air scattered by a centrifugal force caused by the rotation can pass. Further, in the nano-mist generator of the related art, an air guide tube which is installed at an interval on an outer periphery of a counter body and guides the outside air flowing in from above the rotation member to an outer periphery of the porous body is provided, and a pair of large flow passage and small flow passage are formed between the air guide tube and the porous body. Accordingly, a pressure difference is generated in an air flow to generate a turbulence, and a large amount of negative ions and nano-mist are generated.

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CITATION LIST

Patent Literature

- 5 PTL 1: Japanese Patent No. 4016934
PTL 2: Japanese Patent No. 5032389

SUMMARY OF INVENTION

Technical Problem

However, since the nano-mist generator of the related art generates nano-sized mist by crushing large water particles, particle sizes of the water particles become large and a particle size distribution of the water particles tends to widen. Further, since the nano-ion mist generated by the electrostatic atomizer of the related art has a strong electric charge and the nano-mist generated by the nano-mist generator of the related art is negatively charged, for example, the water particles may cause suction or repulsion against the human body. Accordingly, the generated water particles may not enter (cannot infiltrate) the inside of a surface of a horny layer of a human body.

Further, the electrostatic atomizer of the related art requires a Peltier element for condensing water in air and a high-voltage power supply for discharging, and there is a concern that the device becomes complicated and large in size. Further, the nano-mist generator of the related art requires a water tank, an air guide tube, a rotation member, or the like, and there is a concern that the device may become large in size.

The present invention is made to solve the above problems. That is, an object of the present invention is to provide a fine water discharging element capable of aligning a particle size distribution and discharging uncharged fine water, a fine water discharge device and a method for manufacturing the fine water discharging element

Solution to Problem

In order to solve the above problem, according to an aspect of the present invention, there is provided a fine water discharging element capable of transitioning between an adsorption state where water contained in a fluid to be treated is adsorbed and a discharge state where the adsorbed water is discharged to the fluid to be treated. The fine water discharging element comprises a base material portion, a plurality of particles having a core-shell structure including a core portion forming a nucleus and a shell portion formed of a polymer material having a polar functional group capable of hydrogen bonding to cover the core portion and a nanochannel formed between the shell portions of the plurality of particles by laminating the plurality of particles on an outer surface of the base material portion in a film shape, wherein the fine water discharging element is transitioned between the adsorption state and the discharged state by changing the temperature of the water in the nanochannel by controlling an electrifying of at least one of the base material portion and the plurality of particles.

According to this, the plurality of particles is laminated in a film shape on the outer surface of the base material portion formed in a honeycomb shape to form the fine water generating unit. When the plurality of particles is laminated, a nanometer-sized nanochannel is formed between the particles, more specifically, between the shell portions of the particles. Since many polar functional groups are distributed in this nanochannel, the water adsorbed by the particles in

the adsorption state associated with the discharge preparation mode can be quickly moved toward the base material portion. Meanwhile, the adsorbed water quickly moves through the nanochannel in the discharge state associated with the discharge mode and is discharged into the air. In this case, since the nanochannel is formed by the plurality of laminated particles, a large number of openings for discharging the water corresponding to the nanochannel can exist. Accordingly, a particle size distribution of the fine water discharged from the fine water discharge device can be aligned to be 50 nanometers or less.

(Fine Water Discharge Device)

In order to solve the above problem, according to another aspect of the present invention, there is provided a fine water discharge device including a fine water generating unit formed by using the fine water discharging element, a flow path arranged in the fine water generating unit and allowing a first space and a second space to communicate with each other, an air blowing unit introducing air in the first space into the flow path and discharging the air introduced into the flow path into the second space, an electrifying portion electrically connected to the base material portion of the fine water generating unit to electrify the base material portion and a control portion which controls the air blowing unit and the electrifying portion. The control portion includes a discharge control portion which executes a discharge mode of introducing the air in the first space into the flow path by the air blowing unit, electrifying the base material portion of the fine water generating unit by the electrifying portion to discharge the water adsorbed on the particles of the fine water generating unit to the air introduced into the flow path and discharging the fine water to the second space together with the air introduced into the flow path by the air blowing unit and a discharge preparation control portion which executes a discharge preparation mode of introducing the air in the first space into the flow path by the air blowing unit and non-electrifying the base material portion by the electrifying portion to adsorb the water introduced into the flow path to the particles.

Further, in the fine water discharge device, the discharge control portion executes the discharge mode so that the electrifying portion electrifies the base material portion. As a result, temperatures of the particles increase, and the fine water can be discharged. Therefore, the fine water discharge device can discharge a large amount of uncharged fine water into the second space without applying an electric charge to the fine water. Accordingly, the fine water discharged from the fine water discharge device is uncharged and has a small particle size of 50 nanometers or less, and thus, the fine water easily enters (infiltrates) a keratin of a human body, for example. As a result, the fine water can infiltrate the inside of the keratin and can contribute satisfactorily to moisturizing of a skin, for example.

Further, in the fine water discharge device, the discharge preparation control portion executes the discharge preparation mode so that the fine water generating unit (more specifically, particles) can adsorb water in air. Accordingly, in order to adsorb water and discharge fine water, for example, it is not necessary to provide a water tank, a high-voltage power supply, an air guide tube, or the like, and it is possible to reduce a size of the fine water discharge device.

(Manufacturing Method for Fine Water Discharging Element)

In order to solve the above problem, according to a further aspect of the present invention, there is provided a method for manufacturing a fine water discharging element which

adsorbs water contained in a fluid to be treated under a non-electrified state and discharges the adsorbed water to the fluid to be treated under an electrified state, wherein the method comprising the steps of applying a dispersion liquid in which a plurality of particles having a core-shell structure including a core portion forming a nucleus and a shell portion formed of a polymer material having a polar functional group capable of hydrogen bonding to cover the core portion is dispersed to a base material portion formed of a conductive material and drying the particles applied to the base material portion after applying step. Thereafter, the plurality of particles is laminated on an outer surface of the base material portion in a film shape to form a nanochannel between the shell portions of the plurality of particles. By applying the particles having the core-shell structure on the outer surface of the base material portion according to the method above, the fine water discharging element can be manufactured in an extremely simple manner.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a cross-sectional view illustrating a configuration of a fine water discharge device according to an embodiment of the present invention.

FIG. 2 is a partial cross-sectional view illustrating a fine water discharging element constituting the fine water generating unit of FIG. 1.

FIG. 3 is a cross-sectional view illustrating a composition of particles of FIG. 2.

FIG. 4 is a diagram illustrating a measurement result of particle sizes of water particles discharged from the fine water discharging element of FIG. 2.

FIG. 5 is a diagram illustrating a measurement result of the particle size of the water particles after the water particles are discharged from the fine water discharging element of FIG. 2 and a predetermined time has elapsed.

FIG. 6 is a diagram illustrating a measurement result of the particle size of the water particles in indoor air where the water particles are discharged from the fine water discharging element of FIG. 2.

FIG. 7 is a graph illustrating the humidity dependence of dynamic viscoelastic property of the fine water discharging element of FIG. 2.

FIG. 8 is a graph illustrating a temperature dependence of the dynamic viscoelastic property of the fine water discharging element of FIG. 2.

FIG. 9 is a perspective view illustrating a configuration of the fine water generating unit of FIG. 1.

FIG. 10 is a cross-sectional view illustrating configurations of a base material portion and a film portion (particles) constituting the fine water generating unit of FIG. 9.

FIG. 11 is a block diagram of a controller of FIG. 1.

FIG. 12 is a diagram illustrating a measurement result of a time change rate of a water content of a keratin in a case where the fine water is discharged by the fine water discharge device of FIG. 1 and a case where water particles are discharged by a device of the related art.

FIG. 13 is a cross-sectional view illustrating a first modification example of a fine water generating unit according to a second modification example of the embodiment.

FIG. 14 is a cross-sectional view illustrating a second modification example of the fine water generating unit according to the second modification example of the embodiment.

FIG. 15 is a perspective view illustrating a configuration of a fine water generating unit according to a third modification example of the embodiment.

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FIG. 16 is a plan view illustrating a configuration of a fine water generating unit according to a fourth modification example of the embodiment.

FIG. 17 is a cross-sectional view illustrating a detailed configuration of a fine water generating unit of FIG. 16.

FIG. 18 is a cross-sectional view illustrating a first modification example of a fine water generating unit according to a fourth modification example of the embodiment.

FIG. 19 is a cross-sectional view illustrating a second modification example of the fine water generating unit according to the fourth modification example of the embodiment.

FIG. 20 is a cross-sectional view illustrating a third modification example of the fine water generating unit according to the fourth modification example of the embodiment.

FIG. 21 is a cross-sectional view illustrating a configuration of a fine water discharge device according to a sixth modification example of the embodiment.

FIG. 22 is a plan view illustrating a configuration of a fine water generating unit according to a sixth modification example of the embodiment.

FIG. 23 is a plan view illustrating a configuration of a fine water generating unit according to a sixth modification example of the embodiment.

DESCRIPTION OF EMBODIMENTS

Hereinafter, embodiments of the fine water discharge device according to the present invention will be described. For example, a fine water discharge device 100 can be applied to beauty equipment (specifically, moisturizer, booster, vehicle mountable beauty equipment, or the like). As illustrated in FIG. 1, the fine water discharge device 100 includes a fine water discharging element 10 (fine water generating unit 130 described later). The fine water discharging element 10 is capable of transitioning between an adsorption state in which water contained in air is adsorbed and a discharge state in which the adsorbed water is discharged into air. As illustrated in FIG. 2, the fine water discharging element 10 is formed in a sheet shape. The fine water discharging element 10 includes a base material portion 11 and a conductive polymer film portion 12 (hereinafter, simply referred to as a “film portion 12”) which is formed in a film shape on an outer surface of the base material portion 11.

For example, the base material portion 11 is a metal foil formed of stainless steel which is a stainless-steel based metal out of a metal material such as the stainless-steel based metal or a copper-based metal, a carbon material (carbon paper, graphite, or the like), a conductive ceramic material (for example, ITO, or the like), and a conductive resin material (for example, metal-deposited film, nano-silver coating, CNT coating, or the like), which are conductive materials, and has conductivity. As will be described later, the base material portion 11 generates heat when the base material portion 11 is electrified.

The film portion 12 is formed in a film shape by dispersing particles 13 having a core-shell structure in a solvent, applying (application step) a dispersion liquid in which the particles 13 are dispersed to the outer surface of the base material portion 11, and then drying (drying step) the applied dispersion liquid. That is, the fine water discharging element 10 is manufactured according to a manufacturing method including the application step of applying the dispersion liquid in which the particles 13 are dispersed to the conductive base material portion 11 and the drying step of drying

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the particles 13 applied to the base material portion 11 after the application step. As conceptually illustrated in FIG. 2, according to the manufacturing method, the plurality of particles 13 are formed in a layer shape (film shape) to have a certain degree of regularity (most densely) and to be laminated with a plurality of stages (four stages in FIG. 2). Here, regarding characteristics and components of the dispersion liquid, a solvent is water, a solid content concentration is 1 to 3(%), a viscosity is 10 to 200 (mPa·s), pH is 1 to 3, and antifungal agent is added as an additive. Moreover, a drying temperature in the drying step is 100° C. or less.

The particles 13 transition from the discharge state to the adsorption state when the base material portion 11 is not electrified (that is, a state where heat is not generated), and the particles 13 transition from the adsorption state to the discharge state when the base material portion 11 is electrified (that is, a state where heat is generated). Particle size of the particles 13 in the present embodiment is set to about 5 nanometers. Accordingly, the plurality of particles 13 are laminated with a certain degree of regularity on the outer surface of the base material portion 11 to form the film portion 12. Moreover, the “particle size” described in the present specification and claims is an average value of particle sizes of a plurality of randomly selected particles or a plurality of particles existing in a unit volume or a unit area.

Here, the film portion 12 is formed so that a thickness (film thickness) when the plurality of particles 13 are laminated on the outer surface of the base material portion 11 is 1 to 30 micrometers. The particles 13 in the present embodiment are formed of a conductive polymer, and specifically, PEDOT/PSS (poly (3,4-ethylenedioxythiophene)-poly (styrene sulfonic acid)), which is one of the thiophene-based conductive polymers. Therefore, the film portion 12 in the present embodiment is formed to include the particles 13 formed of PEDOT/PSS, and thus, is a thiophene-based conductive polymer.

As illustrated in FIG. 3, the particle 13 includes a core portion 13a and a shell portion 13b. The core portion 13a of the particle 13 forms a nucleus of the particle, and, for example, is formed of poly (3,4-ethylenedioxythiophene), that is, PEDOT, which is a polymer material out of the polymer material or an inorganic material. The shell portion 13b of the particle 13 is formed of a polymer material having a polar functional group 13b1 capable of hydrogen bonding and covers the core portion 13a.

Specifically, the shell portion 13b is formed of poly (styrene sulfonic acid), that is, PSS, which is polystyrene sulfonic acid out of polystyrene sulfonic acid, polyvinyl sulfonic acid, polyvinyl pyrrolidone, polyvinyl alcohol, and sodium polyacrylate. Here, the PSS forming the shell portion 13b includes a sulfonic acid group ($-\text{SO}_3\text{H}$) which is at least one of a sulfonic acid group ($-\text{SO}_3\text{H}$), a carboxyl group ($-\text{COOH}$), a hydroxyl group ($-\text{OH}$), an amino group ($-\text{NH}_2$), an amid group ($-\text{C}=\text{ONH}-$), and a pyrrolidone group (NCOC_3H_6), which are the polar functional groups 13b1 capable of hydrogen bonding.

Here, the PEDOT/PSS which is the particle 13 is set so that a weight ratio of ethylenedioxythiophene (EDOT) which is a monomer of PEDOT and PSS is 1:3 to 1:10, more preferably 1:4 to 1:6.

When the base material portion 11 is not electrified, an amount of water adsorbed by the film portion 12, that is, the particles 13 (shell portion 13b) increases with the elapse of time, and the fine water discharging element 10 is stabilized in a state of a saturated water absorption rate. Further, when the base material portion 11 is electrified and a temperature

of the film portion 12, that is, the particles 13 increases to be equal to or more than a discharge temperature, the fine water discharging element 10 discharges the water bonded to the polar functional group 13b1 of the shell portion 13b of the particles 13 as uncharged (non-electrified) water particles (fine water described later). Here, as a film thickness of the film portion 12 provided in the base material portion 11 decreases, the number of particles 13 which are the conductive polymers supported by the base material portion 11 relatively decreases, and thus, the amount of water adsorbed decreases relatively. Meanwhile, as the film thickness of the film portion 12 provided in the base material portion 11 increases, the number of particles 13 which are the conductive polymers supported by the base material portion 11 relatively increases, and thus, the amount of water adsorbed increases relatively.

Then, in the fine water discharging element 10 configured to include the particles 13 having a core-shell structure, an adsorption speed of water in air in the adsorption state and a discharge speed of water into air in the discharge state are faster than in a general moisture absorbent such as silica gel. Further, in the fine water discharging element 10 configured to include the particles 13 having a core-shell structure, the particle size of the water particles of the water discharged in the discharge state is distributed within a very small particle size distribution range of 2 nanometers to 50 nanometers. Hereinafter, these things will be described in order.

First, the adsorption speed and the discharge speed of water will be described. In the fine water discharging element 10, the particles 13 formed of PEDOT/PSS are laminated on the base material portion 11. In this case, as illustrated in FIG. 2, the particles 13 are laminated in an aligned state in the film portion 12. Further, the PSS constituting the shell portion 13b of the particles 13 has many sulfonic acid groups ($-\text{SO}_3\text{H}$) which are the polar functional groups 13b1 capable of hydrogen bonding, and as illustrated in FIG. 3, has sulfonic acid groups ($-\text{SO}_3\text{H}$) which are the many polar functional groups 13b1 around the shell portion 13b. Therefore, water contained in the air is bonded to the sulfonic acid group ($-\text{SO}_3\text{H}$) which is in contact with the air by hydrogen bond, for example, as bound water having a particle size of 2 nanometers or less.

In this case, when the amount of water on the surface of the film portion 12 is large and the amount of water inside the film portion 12 is small, the shell portion 13b of the particles 13 attempts to move the adsorbed water from the surface of the film portion 12 toward an inside of the film portion 12 by using a difference in water concentration as a driving source. Meanwhile, as illustrated by thick arrows in FIG. 2, inside the film portion 12, a channel 13c (hereinafter, this channel 13c is referred to as a "nanochannel 13c") having a nanometer-sized flow path diameter (for example, about 2 nanometers) is formed between the shell portion 13b and the shell portion 13b of the plurality of laminated particles 13. As illustrated in FIG. 2, the nanochannels 13c are connected to each other inside the film portion 12. Moreover, the flow path diameter of the nanochannel 13c represents a value obtained by averaging the flow path diameters of a plurality of randomly selected nanochannels 13c, or a plurality of nanochannels 13c existing in a unit volume or a unit area or the like.

Many sulfonic acid groups ($-\text{SO}_3\text{H}$), which are polar functional groups 13b1, are distributed in the nanochannel 13c, and water molecules (water) existing on the surface of the film portion 12 move at a high speed to the inside of the film portion 12 through the sulfonic acid groups ($-\text{SO}_3\text{H}$) existing in the nanochannel 13c. As a result, the water in the

air existing on the surface of the film portion 12 moves at a high speed to the inside of the film portion 12 due to the difference in water concentration, and a large amount of water can be adsorbed and retained from the air at a high speed. Here, the water retained inside the film portion 12 is bound water or bounded water, and free water can hardly exist.

Further, since the water moves by using the difference in water concentration as a driving source in this way, when the amount of water inside the film portion 12 is large and the amount of water on the surface of the film portion 12 is small, that is, when air is dried, contrary to the above-mentioned adsorption, the sulfonic acid group ($-\text{SO}_3\text{H}$), which is a polar functional group 13b1 existing in the nanochannel 13c, is transmitted, and thus, the retained (adsorbed) water moves at a high speed from the inside of the film portion 12 toward the surface of the film portion 12. As a result, the water retained (adsorbed) inside the film portion 12 moves at a high speed to the surface of the film portion 12 due to the difference in water concentration, and the film portion 12 (particles 13) can discharge a large amount of water into the air at a high speed as the water particles as the bound water. Further, even when humidity of outside air is high, a temperature of the film portion 12 increases by electrifying the base material portion 11, the film portion 12 (particles 13) can discharge a large amount of water into the air at a high speed as the water particles as the bound water.

Next, it will be described that the water particle size of the water discharged from the film portion 12 (particles 13) is small in the discharge state. In the following, particularly, water having a small water particle size, specifically, water (water cluster) having a particle size of 50 nanometers or less is referred to as "fine water". As illustrated in FIG. 4, a water particle size distribution immediately after being discharged from the fine water discharging element 10 configured to include the particles 13 formed of PEDOT/PSS illustrates the maximum value at 2 nanometers which is a lower limit of a measurement range and is up to around 10 nanometers. Further, as illustrated in FIG. 5, after a predetermined time (for example, 40 seconds) has elapsed from the discharge of water, the water particles (fine water) aggregate (cluster) and the water particle size distribution tends to move to a large particle size side (50 nanometers or less). A peak value of the water particle size distribution is 10 nanometers or less as illustrated in FIGS. 4 and 5. Here, in this measurement, as illustrated in FIG. 6, nanometer-sized water particles are hardly measured in indoor air. Moreover, in the measurement of the water particle size distribution, the measurement is performed using a scanning mobility particle sizer (SMPS) of measuring the particle distribution in a submicron region by classifying the water particles (fine water) using electrical mobility and measuring a number concentration per unit volume (1 cc) using a condensed particle counter.

It is considered that the reason why the fine water discharging element 10 discharges the water having a small water particle size, that is, discharges the fine water, is as follows. As described above, the particles 13 having the core-shell structure in the present embodiment are PEDOT/PSS, and PEDOT is present in the core portion 13a and PSS exists in the shell portion 13b. PSS has a sulfonic acid group ($-\text{SO}_3\text{H}$) which is a hydrophilic group as the polar functional group 13b1, and is distributed in large numbers on the outer surface and the inside of the shell portion 13b.

Therefore, as described above, a large amount of bound water bonded to the sulfonic acid group ($-\text{SO}_3\text{H}$) exists

inside the nanochannel **13c**. Here, it is considered that the reason why the bound water is unevenly distributed in the nanochannels **13c** on the outer surface of the shell portion **13b** and the bound water does not enter the inside of the shell portion **13b** is that PEDOT of the core portion **13a** pushes the bound water to the outer surface of the shell portion **13b** due to hydrophobicity of PEDOT.

Moreover, the reason why the bound water having a size of several nanometers (specifically, the particle size is 2 nanometers) is discharged when the fine water is generated is that, when the base material portion **11** is electrified, the temperature of the film portion **12** (particles **13**) increases and thermal energy is supplied to the bound water (more specifically, water molecules). As a result, in the fine water discharging element **10**, motility of the bound water increases. As a result, the bound water aggregates (clusters) at a spout **13c1** (specifically, a pore having diameter of about 2 nanometers), which is an opening in which a large number of bound water existing in the nanochannel **13c** exists on the surface of the film portion **12** corresponding to the flow path diameter of the nanochannel **13c**, and is discharged into the air as the fine water which is uncharged (non-electrified) and is 50 nanometers or less. In addition, the bound water keeps a shape of the bound water and is discharged as the fine water.

Further, the fine water discharging element **10** is rich in viscoelasticity due to a network between the particles **13** formed to have a core-shell structure from PEDOT/PSS, and stably exists even at high temperature and high humidity. As a result, the fine water discharging element **10** has performance which can be applied to an actual usage environment as described later.

Specifically, in the dynamic viscoelastic property of the film portion **12** including the particles **13**, regarding humidity dependence, as illustrated in FIG. 7, as described above, adsorption (moisture absorption) progresses as humidity increases, elongation indicated by a solid line increases, but viscosity indicated by a broken line and elasticity indicated by a dashed line are appropriately maintained. In the dynamic viscoelastic property of the film portion **12** including the particles **13**, regarding temperature dependence, as illustrated in FIG. 8, the viscosity indicated by a broken line is appropriately maintained although peaks occur at a temperature **T1**, a temperature **T2**, and a temperature **T3** depending on the particles **13** having the core-shell structure, and the elasticity indicated by a dashed line is also appropriately maintained.

Moreover, the temperature **T1** is a 13 relaxation temperature (for example, -60°C.) of the sulfonic acid group ($-\text{SO}_3\text{H}$) constituting the particles **13**. Further, the temperature **T2** is a glass transition temperature (for example, 60°C.) of the shell portion **13b** (PSS) constituting the particles **13**. Further, the temperature **T3** is a melting point (for example, 260°C.) of the core portion **13a** (PEDOT) constituting the particles **13**.

Returning to FIG. 1, the fine water discharge device **100** includes a case **110**, a blower **120** which is an air blowing unit, the fine water generating unit **130**, an electrifying portion **140**, and a controller **150** which is a control portion.

The case **110** is formed in a tubular shape in which both ends are open and extend in a front-rear direction so as to have a flow path **L** through which air flows inside the case **110**. The flow path **L** is open toward a first space **S1** which is a space in front of the case **110** and is open toward a second space **S2** which is a space behind the case **110**. A filter **111** and a filter **112** having air permeability are mounted on the opening of a first space **S1** side and the

opening of a second space **S2** side, respectively. For example, the first space **S1** and the second space **S2** are spaces in the same room, and a user receives a supply of the fine water on the second space **S2** side.

The first space **S1** and the second space **S2** communicate with each other through the flow path **L**. Inside the flow path **L**, the blower **120** and the fine water generating unit **130** (fine water discharging element **10**) are disposed in this order from the first space **S1** side toward the second space **S2** side of the case **110**.

The blower **120** which is the air blowing unit is rotationally driven in a first direction to introduce air in the first space **S1** into the flow path **L**, and discharges the air introduced into the flow path **L** to the second space **S2**. Further, the blower **120** rotationally driven in a second direction opposite to the first direction to introduce the air in the second space **S2** into the flow path **L**, and discharges the air introduced into the flow path **L** to the first space **S1**. For example, the blower **120** is a shaft blower. The blower **120** is electrically connected to the controller **150** and is driven according to a control command value transmitted from the controller **150**. Moreover, since the blower **120** is PWM controlled, the control command value is calculated by a duty ratio.

The fine water generating unit **130** generates and discharges fine water into the air introduced into the flow path **L** by the blower **120**, and adsorbs the water of the air. The fine water generating unit **130** is configured to include the above-mentioned fine water discharging element **10**. Accordingly, in the fine water generating unit **130**, when the base material portion **11** of the fine water discharging element **10** is not electrified, the state of the particles **13** having the core-shell structure constituting the film portion **12** is transitioned, and the particles **13** adsorb (adsorption state) the water of the air introduced into the flow path **L**. Further, in the fine water generating unit **130**, when the base material portion **11** of the fine water discharging element **10** is electrified, the state of the particles **13** is transitioned, and the water adsorbed by the particles **13** is discharged as the fine water to the air introduced into the flow path **L** (discharge state).

As illustrated in FIG. 9, the fine water generating unit **130** includes a columnar honeycomb member **131** having conductivity. The honeycomb member **131** constitutes the base material portion **11** of the fine water discharging element **10**, and is formed of a conductive material, for example, 20 chromium 5 aluminum-based ferritic stainless steel. As illustrated in FIG. 10, the honeycomb member **131** (that is, the base material portion **11**) is formed in a honeycomb shape including a corrugated plate and a flat plate. In the honeycomb member **131** (base material portion **11**), the corrugated plate and the flat plate are insulation-processed. As illustrated in FIG. 10, the particles **13** having a core-shell structure are applied to the outer surface of the honeycomb-shaped base material portion **11** (that is, the honeycomb member **131**) to form the film portion **12**.

Further, as illustrated in FIGS. 1 and 9, the fine water generating unit **130** includes an annular first case member **132** and an annular second case member **133** which surround an outer peripheral portion of the honeycomb member **131** (base material portion **11**). For example, the first case member **132** is disposed on the first space **S1** side of the case **110**, and is reinforced by a flange provided so as to abut on the first space **S1** side of the honeycomb member **131** (base material portion **11**). For example, the second case member **133** is disposed on the second space **S2** side of the case **110**

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and is reinforced by a flange provided so as to abut on the second space S2 side of the honeycomb member 131 (base material portion 11).

Further, in the fine water generating unit 130, an electrode 134 is formed to protrude outward from the honeycomb member 131, that is, the base material portion 11 formed in a honeycomb shape, as illustrated in FIGS. 1 and 9. More specifically, one end of the electrode 134 is electrically connected to the honeycomb member 131 (base material portion 11), and the other end thereof protrudes outward from openings of the first case member 132 and the second case member 133. The electrifying portion 140 is electrically connected to the other end of the electrode 134.

The fine water generating unit 130 including the honeycomb member 131, the first case member 132, the second case member 133, and the electrode 134 is formed as a cartridge that can be detachably assembled to the flow path L of the case 110. In a state where the fine water generating unit 130 to be attached and detached as a cartridge is assembled to the flow path L, the air introduced into the flow path L flows through the fine water generating unit 130 in a direction of an arrow as illustrated by the arrow in FIG. 9.

As illustrated in FIG. 1, the electrifying portion 140 is electrically connected to the electrode 134 of the fine water generating unit 130, and electrifies the fine water generating unit 130 (more specifically, the honeycomb member 131, and the base material portion 11 of the fine water discharging element 10). As illustrated in FIG. 1, the electrifying portion 140 includes electric wires 141, a power supply 142, and a switch 143.

The electric wire 141 is an electric wire which electrically connects the power supply 142, the blower 120, and the electrode 134 of the fine water generating unit 130 to each other. For example, the power supply 142 is a commercial power supply, and can adopt a USB (Universal Serial Bus) power supply system. The switch 143 is arranged on the electric wire 141, and is, for example, a normally open type switch which opens a circuit when is not operated.

The switch 143 closes a circuit according to the control command value transmitted from the controller 150 to electrically connect the power supply 142, the blower 120, and the fine water generating unit 130 to each other, and opens the circuit to cut off the electrical connection between the power supply 142, the blower 120, and the fine water generating unit 130. As a result, the blower 120 is switched between an electrified state in which the blower 120 is rotationally driven in the first or second direction and a non-electrified state in which the rotational driving is stopped. Further, the fine water generating unit 130 is switched between an electrified state in which the honeycomb member 131 (base material portion 11) is electrified to generate the fine water and a non-electrified state in which the honeycomb member 131 (base material portion 11) is not electrified and does not generate the fine water.

As illustrated in FIG. 11, the controller 150 which is the control portion includes a discharge control portion 151 and a discharge preparation control portion 152. The discharge control portion 151 executes a discharge mode which is an operation mode of the fine water discharge device 100, and controls the discharge of the fine water to the second space S2. The discharge preparation control portion 152 executes the discharge preparation mode which is the operation mode of the fine water discharge device 100, and controls a discharge preparation of the fine water to the second space S2.

The discharge mode executed by the discharge control portion 151 is an operation mode of causing the fine water

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generating unit 130 (fine water discharging element 10) to generate the fine water. In addition, the discharge mode executed by the discharge control portion 151 is an operation mode of discharging the air, which is introduced into the flow path L from the first space S1 and includes the fine water, to the second space S2. Specifically, the discharge control portion 151 rotationally drives the blower 120 in the first direction by executing the discharge mode, introduces the air in the first space S1 into the flow path L, and discharges the air in the second space S2.

Then, the discharge control portion 151 closes the switch 143 of the electrifying portion 140 so as to electrify the fine water generating unit 130 (fine water discharging element 10). As a result, the honeycomb member 131 (base material portion 11) of the fine water generating unit 130 (fine water discharging element 10) is electrified. Accordingly, the temperature of the film portion 12 (particles 13) formed on the honeycomb member 131 (base material portion 11) increases to generate the fine water, and the fine water is discharged to the second space S2 together with the air introduced into the flow path L.

Meanwhile, the temperatures of the electrified honeycomb member 131 (base material portion 11) and film portion 12 (particles 13) increase by a range of 20° C. to 50° C. as compared with those of the non-electrified honeycomb member 131 and film portion 12. As a result, the bound water having a particle size of about 2 nanometers adsorbed to the particles 13 passes through the nanochannel 13c and condenses (clusters) and is discharged into the flow path L as the fine water having a particle size of 50 nanometers or less (discharge state), and the air introduced into the flow path L from the first space S1 is discharged from the second space S2 in a state of including the fine water. Moreover, in the discharge mode, first, the discharge control portion 151 electrifies only the honeycomb member 131 (base material portion 11) to increase the temperature of the film portion 12 (particles 13), and then can rotationally drive the blower 120 in the first direction while continuously executing the electrification on the honeycomb member 131 (base material portion 11).

Here, for example, regarding an amount of rotational drive of the blower 120, an amount of electricity applied to the fine water generating unit 130 (fine water discharging element 10), a surface area of the film portion 12 supported by the honeycomb member 131 (base material portion 11), or the like is set as follows. For example, assuming that a temperature of air in the first space S1 is 25° C. and a relative humidity of the air in the first space S1 is 40%, the fine water is included in the air introduced from the first space S1 into the flow path L, and the relative humidity of the air discharged from the second space S2 is set to be about 90%.

The discharge preparation mode executed by the discharge preparation control portion 152 is a mode (operation mode) of causing the fine water generating unit 130 (fine water discharging element 10) to stop the generation of the fine water. In addition, the discharge preparation mode executed by the discharge preparation control portion 152 is a mode (operation mode) of discharging air from the first space S1 toward the second space S2 or from the second space S2 toward the first space S1 via the flow path L.

Specifically, the discharge preparation control portion 152 opens the switch 143 of the electrifying portion 140 so as to switch the fine water generating unit 130 (fine water discharging element 10) to the non-electrified state. Accordingly, the honeycomb member 131 (base material portion 11) of the fine water generating unit 130 (fine water discharging element 10) does not generate heat because the fine

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water generating unit **130** is in the non-electrified state, and as a result, the temperature of the film portion **12** (particles **13**) formed on the honeycomb member **131** (base material portion **11**) decreases.

Further, the discharge preparation control portion **152** rotationally drives the blower **120** in the first direction to discharge the air in the first space **S1** into the second space **S2**. Alternatively, the discharge preparation control portion **152** rotationally drives the blower **120** in the second direction to discharge the air in the second space **S2** into the first space **S1**. Accordingly, when the fine water generating unit **130** (fine water discharging element **10**) is in the non-electrified state, the film portion **12** (particles **13**) formed in the honeycomb member **131** (base material portion **11**) is cooled by the air flowing through the inside of the flow path **L**.

Meanwhile, in a state where the film portion **12** (particle **13**) is cooled, as described above, the water of the air introduced into the flow path **L** is adsorbed by the sulfonic acid group ($-\text{SO}_3\text{H}$) which is the polar functional group **13b1** unevenly distributed in the nanochannel **13c** (adsorption state). In this case, in the non-electrified state, thermal energy is not applied to the water (bound water) bonded to the sulfonic acid group ($-\text{SO}_3\text{H}$) which is the polar functional group **13b1**, and as a result, the water (bound water) does not fly out of the nanochannel **13c** as the fine water, and the water is stably adsorbed.

The fine water discharge device **100** of the present embodiment starts the operation when the user activates the controller **150**, that is, turns on power. Specifically, when the controller **150** is activated, first, as a first step, the discharge preparation control portion **152** executes the discharge preparation mode, rotationally drives the blower **120** in the second direction, and introduces air from the fine water generating unit **130** toward the blower **120** inside the flow path **L**. In this case, the discharge preparation control portion **152** operates the blower **120** so that an operating time is **t1** (s) and a wind speed is **f1** (m/s).

In this way, in the first step, when the discharge preparation control portion **152** executes the discharge preparation mode, the water contained in the air flowing inside the flow path **L** from the second space **S2** to the first space **S1** is adsorbed (moisture-absorbed) by the film portion **12** (particles **13**) supported by the honeycomb member **131** (base material portion **11**). Then, when the discharge preparation control portion **152** operates the blower **120** at the wind speed **f1** (m/s) for the operating time **t1** (s), the blower **120** is stopped.

Subsequently, as a second step, the discharge control portion **151** executes the discharge mode. In the present embodiment, the discharge control portion **151** closes the switch **143** of the electrifying portion **140** and electrifies the honeycomb member **131** (base material portion **11**) of the fine water generating unit **130**. In this case, the discharge control portion **151** electrifies the honeycomb member **131** (base material portion **11**) so that an electrification time is **t2** (s) and electric power is **P** (w) in order to increase the temperature of the honeycomb member **131** (base material portion **11**) by the range of 20°C . to 50°C . as compared with that of the non-electrified honeycomb member **131**. Moreover, the operating time **t2** (s) and the electric power **P** (w) can be arbitrarily set by the user.

Further, as a third step, the discharge control portion **151** electrifies the honeycomb member **131** (base material portion **11**) and at the same time rotationally drives the blower **120** in the first direction to introduce the air from the blower **120** toward the fine water generating unit **130** inside the flow

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path **L**. In this case, the discharge control portion **151** operates the blower **120** so that the operating time is **t2** (s) and the wind speed is **f2** (m/s). For example, the wind speed **f2** (m/s) can be arbitrarily set by the user according to a discharge distance of the fine water.

As described above, in the second step and the third step, the discharge control portion **151** executes the discharge mode. Accordingly, the temperature of the film portion **12** (particles **13**) supported by the honeycomb member **131** (base material portion **11**) increases by about 30°C . to discharge fine water, and the discharged fine water is discharged to the second space **S2** together with the air flowing inside the flow path **L**. Then, when the discharge control portion **151** electrifies the honeycomb member **131** (base material portion **11**) with the electric power **P** (w) for the electrification time **t2** (s) and operates the blower **120** with the wind speed **f2** (m/s) for the operating time **t2** (s), the electrification to the honeycomb member **131** (base material portion **11**) is cut off and the blower **120** is stopped.

Next, as a fourth step, the discharge preparation control portion **152** executes the discharge preparation mode, rotationally drives the blower **120** in the second direction, and introduces the air from the fine water generating unit **130** toward the blower **120** inside the flow path **L**. In this case, the discharge preparation control portion **152** operates the blower **120** so that the operating time is **t3** (s) and the wind speed is **f3** (m/s). Here, the wind speed **f3** (m/s) in the fourth step is set to a value larger than the wind speed **f1** (m/s) in the first step.

As described above, in the fourth step, the discharge preparation control portion **152** executes the discharge preparation mode. Accordingly, the honeycomb member **131** (base material portion **11**) and the film portion **12** (particles **13**) of which the temperatures have been increased by the electrification in the second step are cooled. Here, the wind speed **f3** (m/s) in the fourth step is larger than the wind speed **f1** (m/s) in the first step. Therefore, the air flows quickly inside the flow path **L**, and as a result, the honeycomb member **131** (base material portion **11**) is quickly cooled. Meanwhile, the wind speed **f1** (m/s) in the first step is smaller than the wind speed **f3** (m/s) in the fourth step. Therefore, the air flows slowly inside the flow path **L**, and as a result, a contact time between the film portion **12** (particles **13**) supported by the honeycomb member **131** (base material portion **11**) and the air becomes relatively long, and thus, the water contained in the air is more reliably adsorbed (moisture-absorbed).

Then, the fine water discharge device **100** sets the above-mentioned first to fourth steps as one cycle, and repeats this cycle about 2 to 10 cycles, for example. Moreover, the number of cycles to be repeated is arbitrarily set by the user.

Meanwhile, the above-mentioned fine water discharge device **100** can discharge the generated fine water in a state of the bound water similar to water (natural moisturizing factor) retained in a keratin of a human body and in an uncharged (non-electrified) state. Therefore, by exposing a skin of the user to the fine water discharged from the fine water discharge device **100**, the user can obtain a moisturizing effect of the skin for a long time. This will be explained based on FIG. **12** which illustrates a comparison result of a time change rate of a water content of a keratin of a cheek in a case where the fine water is discharged by the fine water discharge device **100** and a case where nano-mist is discharged by the device of the related art.

Moreover, regarding the fine water discharged by the fine water discharge device **100** (fine water discharging element **10**), pH of the discharged fine water is 6.8 to 7.0, and even

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when the fine water comes into direct contact with a skin, it is safe because there are no reports of erythema or skin abnormalities. Further, a gas generated when the fine water discharge device **100** (fine water discharging element **10**) discharges the fine water includes only a trace amount of sulfur oxides below the environmental standard and does not affect the environment.

Generally, it is said that a keratinous gap of a human skin is about 50 nanometers. As described above, the fine water discharge device **100**, that is, the fine water discharging element **10**, discharges the fine water of 50 nanometers or less in the uncharged (non-electrified) state and the state of the bound water. As a result, as illustrated by a solid line in FIG. **12**, the fine water discharged from the fine water discharge device **100** (fine water discharging element **10**) efficiently infiltrates the inside of the keratin (inside of the skin) in the state of the bound water, and as a result, a water content of the keratin is unlikely to decrease with a passage of time. Therefore, the fine water can provide a moisturizing effect on the skin for a long time, for example, improvement of dry fine wrinkles, smoothing of a skin texture, improvement of sagging and firmness, suppression of excessive sebum secretion, moisturizing of an atopic skin (dry skin) can be expected.

Meanwhile, the device of the related art discharges nano-sized nano-mist in a charged and free water state. As a result, as illustrated by a dashed line in FIG. **12**, the nano-mist discharged from the device of the related art infiltrate the inside of the keratin (inside of the skin) immediately after reaching the skin. However, the nano-mist infiltrates in the free water state, the nano-mist easily evaporates, and as a result, the water content of the keratin decreases with the passage of time. Therefore, it is difficult for the nano-mists to provide a moisturizing effect on the skin for a long time in the related art.

Moreover, as described above, the fine water is discharged as the temperature of the film portion **12** (particles **13**) in which the fine water is supported by the honeycomb member **131** (base material portion **11**) increases. Generally, it is said that the keratin has a high transdermal absorption rate at 32° C. to 39° C. Therefore, it can be considered that the discharge of the fine water at a warm temperature also facilitates the infiltration to the inside of the keratin (inside of the skin).

Further, depending on the user, a cream or the like (for example, APM (magnesium phosphate ascorbate) or the like) which infiltrates a beauty component such as vitamin C into the keratin (into the skin) may be applied to a cheek or the like. Meanwhile, the fine water discharged by the fine water discharge device **100** (fine water discharging element **10**) is uncharged (non-electrified), and can stably maintain a fine particle shape, that is, 50 nanometers or less even after the fine water is discharged. Therefore, for example, the fine water which has not infiltrated the inside of the keratin (inside of the skin) can be infiltrated together with the beauty component into epidermis of the skin, and as a result, a beauty component infiltration effect of infiltrating the beauty component into the keratin (into the skin) can be enhanced, and for example, skin whitening and skin firmness can be expected.

Meanwhile, the nano-mist discharged by the device of the related art is charged, and it is difficult to maintain a fine particle shape after the discharge. Therefore, for example, the nano-mist which has not infiltrated the inside of the keratin (inside of the skin) cannot infiltrate together with the beauty component into the epidermis of the skin when the

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particle size is larger than 50 nanometers, and as a result, the beauty component infiltration effect cannot be enhanced.

Here, when the fine water discharged from the fine water discharge device **100** to the second space **S2** is supplied to the skin of the user, for example, a discharge-side opening (that is, nozzle) of the flow path **L** communicating with the second space **S2** is formed in a blowout shape which exerts the Coanda effect and the fine water discharged from the case **110** can be directly supplied (touched) to the skin of the user. Further, for example, it is possible to supply the fine water to the skin and infiltrate the fine water through a face mask which covers a face of the user. Further, the second space **S2** is partitioned by a tent, a capsule, or the like, the fine water is discharged from the case **110** into the tent, the capsule, or the like, and the fine water is supplied to the skin of the user who has entered the tent, the capsule, or the like and the fine water can be infiltrated. Furthermore, it is possible to supply the fine water to the skin (scalp) and can infiltrate the fine water through a hat (cap) or helmet worn by the user.

As can be understood from the above description, the fine water discharge device **100** according to the above embodiment includes the case **110** which has the flow path **L** allowing the first space **S1** and the second space **S2** to communicate with each other, the blower **120** which is the air blowing unit which is arranged in the flow path **L**, introduces air in the first space **S1** into the flow path **L**, and discharges the air introduced into the flow path **L** into the second space **S2**, the fine water generating unit **130** which is disposed in the flow path **L**, includes the honeycomb member **131** (base material portion **11**) formed in a honeycomb shape as the base material portion and the plurality of particles **13** having the core-shell structure including the core portion **13a** forming a nucleus and the shell portion **13b** formed of a polymer material having the polar functional group capable of hydrogen bonding to cover the core portion **13a**, and in which at least one of the honeycomb member **131** (base material portion **11**) and the particles **13** has conductivity, the plurality of the particles **13** are laminated as the film-shape conductive polymer film portion **12** on the outer surface of the honeycomb member **131** (base material portion **11**), and states of the particles **13** are transitioned between the adsorption state where water is adsorbed and the discharge state where the adsorbed water is discharged to the air to generate fine water, the electrifying portion **140** electrically connected to the honeycomb member **131** (base material portion **11**) of the fine water generating unit **130** to electrify the honeycomb member **131** (base material portion **11**), and the controller **150** which is the control portion controlling the blower **120** and the electrifying portion **140**, in which the controller **150** includes the discharge control portion **151** which executes the discharge mode of introducing the air in the first space **S1** into the flow path **L** by the blower **120**, electrifying the honeycomb member **131** (base material portion **11**) of the fine water generating unit **130** by the electrifying portion **140** to discharge the water adsorbed on the particles **13** of the fine water generating unit **130** to the air introduced into the flow path **L** as the uncharged fine water having a particle size of 50 nanometers or less through the spout **13c1** which is the opening of the nanochannel **13c** formed by the plurality of laminated particles **13**, and discharging the fine water to the second space **S2** together with the air introduced into the flow path **L** by the blower **120**, and the discharge preparation control portion **152** which executes a discharge preparation mode of introducing the air in the first space **S1** or the second space **S2** into the flow path **L** by the blower **120** and non-electrifying the

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honeycomb member **131** (base material portion **11**) by the electrifying portion **140** to adsorb the water introduced into the flow path **L** to the particles **13** through the nanochannel **13c** from the spout **13c1**.

In this case, the honeycomb member **131** (base material portion **11**) is formed of a stainless-steel based metal, and the particles **13** are PEDOT/PSS including the core portion **13a** formed of poly (3,4-ethylenedioxythiophene) and the shell portion **13b** formed of poly (styrene sulfonic acid), and PEDOT/PSS is set so that a weight ratio of ethylenedioxythiophene (EDOT), which is a monomer of PEDOT, and PSS is between 1:4 and 1:6.

Further, in these cases, the fine water discharged into the second space **S2** in the discharge mode is supplied to a human body existing in the second space **S2**.

Accordingly, the plurality of particles **13** are laminated as the film portion **12** on the outer surface of the honeycomb member **131** (base material portion **11**) formed in a honeycomb shape to configure the fine water generating unit **130**. When the plurality of particles **13** are laminated as the film portion **12**, nanometer-sized nanochannels **13c** are formed between the particles **13**, more specifically, between the shell portions **13b** of the particles **13**. Since many polar functional groups **13b1** are distributed in this nanochannel **13c**, the water adsorbed by the particles **13** in the adsorption state can be quickly moved toward the honeycomb member **131** (base material portion **11**). Meanwhile, the adsorbed water quickly moves through the nanochannel **13c** in the discharge state and is discharged into the air. In this case, since the nanochannel **13c** is formed by a plurality of laminated particles **13**, a large number of spouts **13c1** having a size of about 2 nanometers, which are openings for discharging the water corresponding to the nanochannel **13c**, can exist. Accordingly, the particle size distribution of the water particles discharged from the fine water discharge device **100** can be aligned to be 50 nanometers or less.

Further, in the fine water discharge device **100**, when the discharge control portion **151** executes the discharge mode, the electrifying portion **140** electrifies the honeycomb member **131** (base material portion **11**). As a result, the temperatures of the particles **13** increase, and the fine water can be discharged. Therefore, the fine water discharge device **100** can discharge a large amount of uncharged fine water into the second space **S2** without applying an electric charge to the fine water. Accordingly, the fine water discharged from the fine water discharge device **100** is uncharged and has a small particle size of 50 nanometers or less. Therefore, when the fine water is supplied to a human body existing in the second space **S2**, the fine water easily enters (infiltrates) a keratin of the human body. As a result, the fine water can infiltrate the inside of the keratin and can contribute satisfactorily to moisturizing of a skin or the like.

Further, in the fine water discharge device **100**, the discharge preparation control portion **152** executes the discharge preparation mode so that the fine water generating unit **130** (more specifically, the particles **13**) can adsorb the water of the air. Accordingly, in order to adsorb water and discharge the fine water, for example, it is not necessary to provide a water tank, a high-voltage power supply, an air guide tube, or the like, and it is possible to reduce a size of the fine water discharge device **100**.

In this case, the discharge control portion **151** and the discharge preparation control portion **152** execute the cycle including the discharge mode and the discharge preparation mode a plurality of times.

According to this, for example, the discharge control portion **151** executes the discharge mode on the water which

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is adsorbed (moisture-absorbed) by the particles **13** of the honeycomb member **131** (base material portion **11**) by the discharge preparation control portion **152** executing the discharge preparation mode, and thus, the fine water can be discharged to the second space **S2**. That is, by executing the cycle including the discharge mode and the discharge preparation mode, the water adsorbed (moisture-absorbed) from the air can be discharged as the fine water. Accordingly, it is not necessary to provide a water tank, a high-voltage power supply, an air guide tube, or the like, and it is possible to reduce the size of the fine water discharge device **100**.

Further, since the cycle including the discharge mode and the discharge preparation mode can be executed a plurality of times, for example, the user of the fine water discharge device **100** can execute the cycle an arbitrary number of times. Therefore, the fine water discharge device **100** can be used according to the usage pattern of the user, and a convenience of the user can be improved.

Further, in these cases, in the discharge mode, the temperature of the electrified honeycomb member **131** (base material portion **11**) increases by the range of 20° C. to 50° C. as compared with that of the non-electrified honeycomb material **131**.

According to this, it is possible to reduce the thermal energy required when discharging the fine water into the second space **S2** in the discharge mode. Meanwhile, in the discharge preparation mode, the honeycomb member **131** (base material portion **11**) is in the non-electrified state. Therefore, less energy is required to operate the fine water discharge device **100**, and as a result, energy saving can be achieved.

In these cases, the core portion **13a** is formed of poly (3,4-ethylenedioxythiophene) out of a polymer material or an inorganic material. The shell portion **13b** has, as the polar functional group **13b1**, a sulfonic acid group ($-\text{SO}_3\text{H}$) which is at least one of a sulfonic acid group ($-\text{SO}_3\text{H}$), a carboxyl group ($-\text{COOH}$), a hydroxyl group ($-\text{OH}$), an amino group ($-\text{NH}_2$), an amid group ($-\text{C}=\text{ONH}-$), and a pyrrolidone group (NCOC_3H_6). In this case, the shell portion **13b** is formed of, as the polymer material, polystyrene sulfonic acid (PSS) which is at least one of polystyrene sulfonic acid, polyvinyl sulfonic acid, polyvinyl pyrrolidone, polyvinyl alcohol, and sodium polyacrylate. Further, the honeycomb member **131** (base material portion **11**) is formed of 20 chromium 5 aluminum-based ferritic stainless steel which is stainless steel which is at least one of a metal material such as a stainless-steel based metal or a copper-based metal, a carbon material, a conductive ceramic material, and a conductive resin material.

Accordingly, the particles **13** having a core-shell structure can have a large number of sulfonic acid groups ($-\text{SO}_3\text{H}$) in which the shell portion **13b** formed of PSS is the polar functional group **13b1** capable of hydrogen bonding. Accordingly, the water contained in the air can be quickly and abundantly adsorbed in the adsorption state, and the adsorbed water can be quickly discharged to the air in the discharge state. As a result, the fine water discharge device **100** can discharge the fine water into the air while absorbing (adsorbing) the water from the air. Therefore, the fine water discharge device **100** can discharge the fine water by utilizing the water contained in the air.

First Modification Example

In the above embodiment, the particles **13** forming the film portion **12** have a core-shell structure formed of PEDOT/PSS, which is a conductive polymer material. That

is, in the above embodiment, the core portion **13a** of the particle **13** is formed of PEDOT, and the shell portion **13b** of the particle **13** is formed of PSS.

Instead of this, it is possible to configure the particles **13** having a core-shell structure in which the core portion **13a** of the particle **13** is formed of cerium oxide (CeO) which is an inorganic material and the shell portion **13b** of the particle **13** is formed of polyvinylpyrrolidone, that is, PVP out of polystyrene sulfonic acid, polyvinylsulfonic acid, polyvinylpyrrolidone, polyvinyl alcohol, and sodium polyacrylate, which are polymer materials having the polar functional group **13b1** capable of hydrogen bonding. In this case, the PVP forming the shell portion **13b** has a pyrrolidone group (NCOC₃H₆) out of a sulfonic acid group (—SO₃H), a carboxyl group (—COOH), a hydroxyl group (—OH), an amino group (—NH₂), an amide group (—C=ONH—), and the pyrrolidone group (NCOC₃H₆), which are the polar functional group **13b1** capable of hydrogen bonding.

As described above, even in a case where the particles **13** having the core-shell structure are formed of CeO/PVP and the film portion **12** containing the particles **13** is formed in a layer on the base material portion **11**, when the base material portion **11** is in a non-electrified state, with the passage of time, the amount of water adsorbed by the particles **13** (shell portion **13b**) increases and is stabilized in the state of the saturated water absorption rate. Further, in the fine water discharging element **10**, the base material portion **11** is electrified and the temperatures of the particles **13** increase to be equal to or higher than a discharge temperature, and thus, the water adsorbed on the particles **13** (shell portion **13b**) is discharged.

Then, even in the case of this first modification example, the shell portions **13b** of the particles **13** are laminated in the film portion **12** in an aligned state as illustrated in FIG. 2. Further, the PVP constituting the shell portion **13b** of the particles **13** has a large number of pyrrolidone groups (NCOC₃H₆) which are polar functional groups **13b1** capable of hydrogen bonding, and as illustrated in FIG. 3, has many pyrrolidone groups (NCOC₃H₆) around the shell portion **13b**. Therefore, even in the film portion **12** of this first modification example, many pyrrolidone groups (NCOC₃H₆) are distributed in the nanochannel **13c**, and the water contained in the air is adsorbed by hydrogen bonding on the pyrrolidone group (NCOC₃H₆) which is in contact with the air which is a fluid to be treated.

Accordingly, even in the fine water discharging element **10** of this first modification example, the water adsorbed on the surface of the film portion **12** moves at a high speed to the inside of the film portion **12** due to the difference in water concentration, and thus, a large amount of water can be adsorbed and retained from the air at a high speed. Further, since the water moves by using the difference in water concentration as the driving source in this way, when the amount of water inside the film portion **12** is large and the amount of water on the surface of the film portion **12** is small, that is, when air is dried, contrary to the above-mentioned adsorption, the water retained (adsorbed) on the surface of the film portion **12** moves from the inside of the film portion **12** at a high speed along the pyrrolidone group (NCOC₃H₆) existing in the nanochannel **13c**. As a result, even in the fine water discharging element **10** of the first modification example, the water retained (adsorbed) inside the film portion **12** moves at a high speed to the surface of the film portion **12** due to the difference in water concentration, and thus, a large amount of water can be discharged to the air at a high speed. Therefore, even in this first

modification example, the same effects as those of the above embodiment can be obtained.

Second Modification Example

In the above embodiment, the honeycomb shape of the honeycomb member **131** (base material portion **11**) is triangular. Instead of this, the honeycomb shape of the honeycomb member **131** (base material portion **11**) can be a wave shape (first modification example) as illustrated in FIG. 13 or a hexagonal shape (second modification example) as illustrated in FIG. 14. Even when the honeycomb shape is adopted for the honeycomb member **131** (base material portion **11**), the particles **13** having a core-shell structure are applied to the surface of the honeycomb member **131** (base material portion **11**) to form the film portion **12**.

As a result, even in this case, in the fine water generating unit **130** (fine water discharging element **10**), when the honeycomb member **131** (base material portion **11**) is not electrified, the particles **13** having a core-shell structure constituting the film portion **12** can adsorb the water of the air introduced into the flow path L (adsorption state), and when the honeycomb member **131** (base material portion **11**) is electrified, the water adsorbed by the particles **13** can be discharged as the fine water to the air introduced into the flow path L (discharge state). Therefore, the same effects as those of the above embodiment can be obtained.

Third Modification Example

In the above embodiment, the honeycomb member **131** (base material portion **11**) has a columnar shape. Instead of this, as illustrated in FIG. 15, the honeycomb member **131** (base material portion **11**) can be made prismatic. In this case, as for the honeycomb shape of the honeycomb member **131** (base material portion **11**), the shapes of the first modification example illustrated in FIGS. 10 and 13 and the shape of the second modification example illustrated in FIG. 14 can be adopted.

Even when the prismatic honeycomb member **131** (base material portion **11**) is adopted, the particles **13** having a core-shell structure are applied to the surface of the honeycomb member **131** (base material portion **11**) to form the film portion **12**. As a result, even in this case, in the fine water generating unit **130** (fine water discharging element **10**), when the honeycomb member **131** (base material portion **11**) is not electrified, the particles **13** having a core-shell structure constituting the film portion **12** can adsorb the water of the air introduced into the flow path L (adsorption state), and when the honeycomb member **131** (base material portion **11**) is electrified, the water adsorbed by the particles **13** can be discharged as the fine water to the air introduced into the flow path L (discharge state). Therefore, the same effects as those of the above embodiment can be obtained.

Fourth Modification Example

In the above embodiment, the fine water generating unit **130** includes the honeycomb member **131**, and the base material portion **11** of the fine water discharging element **10** is integrally formed with the honeycomb member **131**. Instead of this, it is also possible to provide a fine water generating unit **230** illustrated in FIG. 16. The fine water generating unit **230** includes the base material portion **11** formed in a spiral cross section using one fine water discharging element **10**, and the particles **13** having a core-shell

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structure are applied to the outer surface (a surface corresponding to the inside of the spiral shape) of the base material portion 11 to form the film portion 12.

In this case, the fine water generating unit 230 has a liner portion 231 and a corrugated portion 232 disposed on the surface side of the film portion 12. Each of the liner portion 231 and the corrugated portion 232 is provided in a spiral cross section. As illustrated in FIG. 17, the liner portion 231 is formed of a metal foil punching plate. The corrugated portion 232 is formed of pulp paper. Further, in this case, the fine water generating unit 230 includes electrodes 233 connected to the electrifying portion 140. The electrodes 233 are provided at both ends of the base material portion 11 of the fine water discharging element 10 constituting the fine water generating unit 230, respectively.

In the fine water generating unit 230 configured in this way, air flows through a gap formed between the corrugated portion 232 and the liner portion 231. Then, in the fine water generating unit 230 (fine water discharging element 10), when the base material portion 11 is not electrified, the particles 13 having a core-shell structure constituting the film portion 12 adsorb the water of the air introduced into the flow path L (adsorption state), and when the base material portion 11 is electrified, the water adsorbed by the particles 13 is discharged to the air introduced into the flow path L (discharge state). Moreover, when the base material portion 11 and the corrugated portion 232 are formed so as to be in contact with each other, the corrugated portion 232 is heated by heat generated when the base material portion 11 is electrified. Then, when the corrugated portion 232 is heated, the liner portion 231 is also heated, and as a result, the film portion 12 of the fine water discharging element 10, that is, the particles 13 are heated and quickly transitioned to the discharge state. Therefore, the same effects as those of the above embodiment can be obtained.

Here, various changes can be made to the liner portion 231 and the corrugated portion 232 constituting the fine water generating unit 230. For example, as illustrated in FIG. 18 as a first modification example, the liner portion 231 can be omitted, and a metal mesh 234 can be formed instead of the corrugated portion 232. Further, as illustrated in FIG. 19 as a second modification example, the liner portion 231 can be omitted and the corrugated portion 232 can be formed of a metal plate. Further, as illustrated in FIG. 20 as a third modification example, the liner portion 231 can be formed of a metal plate and the liner portion 231 can be disposed between the base material portion 11 and the corrugated portion 232. Further, instead of the punching metal, the liner portion 231 can be formed of a metal mesh. Even when the liner portion 231 and the corrugated portion 232 are changed in this way, the same effects as those of the above embodiment can be obtained.

Fifth Modification Example

In the above-described embodiment and each of the above-described modification examples, the user directly operates the controller 150 which can be operated by the user to operate the fine water discharge device 100. In addition to this, for example, it is also possible to operate the fine water discharge device 100 by operating a mobile terminal (not illustrated) (specifically, for example, a smartphone or the like) possessed by the user. In this case, an application for selecting and instructing a control content, that is, an operation mode by the controller 150 is installed in the mobile terminal. Further, in this case, the fine water discharge device 100 includes a communication device 160

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capable of communicating with the mobile terminal and communicating with the controller 150, as illustrated by a broken line in FIG. 1.

When the fine water discharge device 100 is operated by using the mobile terminal, the user selects the operation mode of the fine water discharge device 100 in the application activated in the mobile terminal, and thus, the selected operation mode is transmitted to the communication device 160 of the fine water discharge device 100. In the fine water discharge device 100, when the operation mode is received via the communication device 160, the controller 150 controls the operations of the blower 120 and the electrifying portion 140 according to the operation mode. As a result, the fine water discharge device 100 can be operated according to the operation mode selected by the user using the mobile terminal.

Meanwhile, the application installed on the mobile terminal can have a function of setting an arbitrary operation mode (aspect) in which the above-mentioned discharge mode and discharge preparation mode are combined with each other. Thereby, for example, the user can arbitrarily set a time at which the fine water is discharged, a wind speed by the blower 120, or the like.

In addition, for example, when a skin measurement application for measuring the amount of water in the skin or the like is installed on the mobile terminal, the fine water discharge device 100 can automatically set and operate an optimum operation mode according to a measurement result by the skin measurement application, that is, a skin condition of the user. In this case, the measurement result by the skin measurement application is transmitted to the controller 150 via the communication device 160. Accordingly, for example, when the water content of the skin of the user is small, the controller 150 sets a long time for discharging the fine water (that is, a time in the discharge mode), or sets a large number of times for discharging the fine water (that is, the number of times in the discharge mode). Therefore, the user does not need to set the time and the number of times by himself/herself and can use the fine water discharge device 100 extremely easily and optimally.

Further, in this case, the communication device 160 can transmit data such as an operation content according to the skin measurement application of the fine water discharge device 100, specifically, the time or number of discharge modes according to the water content of the skin to an external server while ensuring confidentiality. Accordingly, by analyzing various data transmitted to the server and stored therein as big data, for example, an analysis result can be used for upgrading the skin measurement application, or for improving a control content of the fine water discharge device 100 by the controller 150. As a result, the user can operate and use the fine water discharge device 100 more finely according to his/her skin condition.

Sixth Modification Example

In the above-described embodiment and each of the above-described modification examples, the fine water discharge device 100 includes the fine water generating unit 130 (fine water discharging element 10) having the single honeycomb member 131, and the single honeycomb member 131 can be switched between the electrified state and the non-electrified state. Instead of this, the fine water discharge device 100 may include the fine water generating unit 130 (fine water discharging element 10) having a plurality of honeycomb members 131, and each honeycomb member 131 can be independently switched between the electrified

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state and the non-electrified state. Hereinafter, this sixth modification example will be specifically described.

In the sixth modification example, for example, the fine water discharge device **100** includes two honeycomb members **131**, as illustrated in FIG. **21**. Moreover, in the following description, when the plurality of honeycomb members **131** are distinguished, the honeycomb members **131** are designated by reference numerals “A”, “B”, For example, when two honeycomb members **131** are distinguished, one honeycomb member **131** is referred to as a honeycomb member **131A**, and the other honeycomb member **131** is referred to as a honeycomb member **131B**.

The honeycomb member **131A** and the honeycomb member **131B** are independently switched between the electrified state and the non-electrified state, respectively. Therefore, as illustrated in FIG. **21**, the electrodes **134** are electrically connected to the honeycomb member **131A** and the honeycomb member **131B**, respectively. Then, each electrode **134** is connected to each of the two switches **143**, that is, provided corresponding to each electrode **134** via the electric wire **141**. Here, in the electrifying portion **140**, the switch **143**, which matches the number of electrodes **134**, in other words, the number of honeycomb members **131**, is provided. For example, the electric wire **141** connected to the electrode **134** electrically connected to the honeycomb member **131A** is accommodated inside a wall portion of the case **110**.

The discharge control portion **151** of the controller **150** electrifies the honeycomb member **131A** in the discharge mode to generate the fine water from the honeycomb member **131A**. Meanwhile, the discharge preparation control portion **152** of the controller **150** non-electrifies the honeycomb member **131B** and adsorbs the water of air to the honeycomb member **131B** in the discharge preparation mode. Then, when a predetermined time elapses, the discharge control portion **151** electrifies the honeycomb member **131B** and generates the fine water from the honeycomb member **131B** in the discharge mode. Meanwhile, the discharge preparation control portion **152** non-electrifies the honeycomb member **131A** and adsorbs the water of the air to the honeycomb member **131A** in the discharge preparation mode. In this way, the discharge control portion **151** and the discharge preparation control portion **152** cooperate with each other, and the electrified states and the non-electrified states of the honeycomb member **131A** and the honeycomb member **131B** are switched repeatedly every time a predetermined time elapses. Accordingly, the fine water discharge device **100** can continuously discharge the fine water, and the same effects as those of the above embodiment can be obtained.

Here, the number (number of divisions) of the honeycomb members **131** is not limited to two as described above. For example, as illustrated in FIG. **22**, it is also possible to provide honeycomb members **131A** to **131H** by executing division into eight in a circumferential direction. Further, for example, as illustrated in FIG. **23**, it is also possible to provide four honeycomb members **131A** to **131D** having a diameter smaller than that of the case **110**. In this way, when two or more honeycomb members **131** are provided, a ratio between the honeycomb member **131** in the electrified state and the honeycomb member **131** in the non-electrified state can be freely changed. Moreover, the shape of the divided honeycomb member **131** may be any shape as long as it can be accommodated inside the case **110**, more specifically, as long as it can be accommodated in the first case member **132** and the second case member **133**.

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In the case illustrated in FIG. **22**, the discharge control portion **151** and the discharge preparation control portion **152** cooperate with each other every time a predetermined time elapses, for example, the electrified states and the non-electrified states of the honeycomb member **131A**, the honeycomb member **131C**, the honeycomb member **131E**, and the honeycomb member **131G** can be switched repeatedly, and the non-electrified states and the electrified states of the honeycomb member **131B**, the honeycomb member **131D**, the honeycomb member **131F**, and the honeycomb member **131H** can be switched repeatedly. Accordingly, in the fine water discharge device **100**, the ratios of the honeycomb member **131** in the electrified state and the honeycomb member **131** in the non-electrified state can be made equal to each other, and the fine water can be continuously discharged.

Similarly, in the case illustrated in FIG. **23**, the discharge control portion **151** and the discharge preparation control portion **152** cooperate with each other every time a predetermined time elapses, for example, the non-electrified states and the electrified states of the honeycomb member **131A** and the honeycomb member **131C** can be switched repeatedly and the non-electrified states and the electrified states of the honeycomb member **131B** and the honeycomb member **131D** can be switched repeatedly. Accordingly, in the fine water discharge device **100**, the ratios of the honeycomb member **131** in the electrified state and the honeycomb member **131** in the non-electrified state can be made equal to each other, and the fine water can be continuously discharged.

By the way, in particular, as illustrated in FIGS. **22** and **23**, when the fine water discharge device **100** has three or more honeycomb members **131**, as the ratio of the honeycomb member **131** in the electrified state and the honeycomb member **131** in the non-electrified state, for example, it is possible to increase the number of honeycomb members **131** which are not electrified as compared to the honeycomb members **131** which are electrified. In this case, the honeycomb member **131** in the non-electrified state increases the time for adsorbing the water in the air, and as a result, the amount of water adsorbed can increase. Therefore, when the honeycomb member **131** is switched to the electrified state, more fine water can be discharged, and the same effects as those of the above embodiment can be obtained.

Further, when the fine water discharge device **100** has a plurality of honeycomb members **131**, the discharge control portion **151** and the discharge preparation control portion **152** can continuously switch between the electrified state and the non-electrified state so that the electrified state and the non-electrified state temporally overlap (wrap) each other between the respective honeycomb members **131**. In this way, the amount of fine water generated by the fine water discharge device **100** can be made uniform by continuously switching between the electrified state and the non-electrified state so that the electrified state and the non-electrified state temporally overlap (wrap) each other, and the same effects as those of the above embodiment can be obtained.

The practice of the present invention is not limited to the above-described embodiment and each of the above-mentioned modification examples, and various modifications can be made without departing from the object of the present invention.

For example, in the above-described embodiment and each of the above-described modification examples, the base material portion **11** is formed of a conductive material. Instead of this, when the particles **13** have conductivity such

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as PEDOT/PSS, the base material portion **11** can be formed of a non-conductive material.

Further, in the above-described embodiment and each of the above-described modification examples, the fine water discharge device **100** is configured to integrally include the blower **120**. However, it is also possible to configure the fine water discharge device **100** to include the blower **120** as a separate body. In this case, the fine water discharge device **100** excluding the blower **120** can be attached to a device having a blower function, for example, a dryer, an air conditioner, a fan, or the like. As a result, the controller **150** of the fine water discharge device **100** electrifies or does not electrify the fine water generating unit **130** (fine water discharging element **10**) in accordance with the fact that the device having the blower function blows air, and thus, the fine water discharge device **100** can discharge fine water or adsorb water. Therefore, even in this case, the same effects as those of the above-described embodiment and the above-described modification examples can be expected.

Further, in the above-described embodiment and each of the above-described modification examples, the discharge preparation control portion **152** constituting the controller **150** executes the discharge preparation mode, and thus, the fine water generating unit **130** (fine water discharging element **10**) adsorbs (moisture-adsorbs) the water in the air introduced into the flow path **L** of the case **110**. Instead of or in addition to this, for example, inside the case **110**, a sheet member moistened with water can be disposed to be interposed between the blower **120** and the fine water generating unit **130** (honeycomb member **131**, first case member **132**, and second case member **133**) inside the flow path **L**. Alternatively, inside the case **110**, a spray mechanism for spraying vapor-like (mist-like) water can be disposed inside the flow path **L** between the blower **120** and the fine water generating unit **130**.

When the sheet member or the spray mechanism is disposed in this way, in the discharge preparation mode, unlike the case of the above embodiment, the discharge preparation control portion **152** rotationally drives the blower **120** in the first direction to flow air from the first space **S1** side to the second space **S2** side in the flow path **L**. Accordingly, the water contained in the sheet member or the water sprayed from the spray mechanism is contained in the air and supplied to the fine water generating unit **130** (fine water discharging element **10**). Therefore, the sheet member or the spray mechanism can effectively assist the film portion **12** (particles **13**) cooled in the non-electrified state adsorbing (moisture-absorbing) water (water clusters).

Further, in the above-described embodiment and each of the above-described modification examples, the fine water discharge device **100** discharges only fine water into the second space **S2**. In addition to this, the fine water discharge device **100** can be configured to discharge steam-like water or chemicals at the same time as the fine water. In this case, for example, the fine water discharge device **100** can include a steam generator having a tank for storing water or chemicals. As a result, as described above, at the same time that the fine water discharge device **100** discharges fine water, the steam generator can steam the water or the chemicals and discharge the steamed water or chemicals to the second space **S2**.

Further, in the above-described embodiment and each of the above-described modification examples, the fine water generating unit **130** is configured so that the film portion **12** (particles **13**) is provided on the entire honeycomb member **131**. Instead of this, for example, a through hole is provided in the central portion of the honeycomb member **131**, a

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fragrance agent such as aroma oil can be stored in the through hole. In this case, the discharge control portion **151** executes the discharge mode to increase the temperature of the honeycomb member **131** (base material portion **11**) in the electrified state. Accordingly, as described above, the fine water can be discharged into the second space **S2**, the stored fragrance agent can be evaporated, and a fragrance component can also be discharged into the second space **S2**.

Further, in the above-described embodiment and each of the above-described modification examples, the fine water discharge device **100** is configured to include one blower **120** and one fine water generating unit **130** inside the case **110**. Instead of this, it is also possible to configure the fine water discharge device **100** to include a plurality of the blowers **120** and a plurality of the fine water generating units **130** inside the case **110**. In this case, for example, a plurality of pairs consisting of the blowers **120** and the fine water generating units **130** can be provided in parallel in a direction orthogonal to an axial direction of the case **110**, or the plurality of pairs can be provided in series along the axial direction of the case **110**.

Accordingly, each pair (blower **120** and fine water generating unit **130**) can be operated in a sequence or at the same time. Therefore, in this case, a discharge amount or a discharge distance of the fine water discharged into the second space **S2** can be adjusted, or a discharge time when the fine water is continuously discharged can be adjusted.

Further, in the above-described embodiment and each of the above-described modification examples, when the controller **150** generates fine water and discharges the fine water, the controller **150** closes the switch **143** of the electrifying portion **140** to operate the blower **120** and electrify the fine water generating unit **130**. Meanwhile, when the controller **150** stops the generation and discharge of the fine water, the controller **150** opens the switch **143** of the electrifying portion **140** to stop the operation of the blower **120** and non-electrify the fine water generating unit **130**.

Instead of this, for example, in a configuration in which the electrification to the blower **120** can be maintained and the operation of the blower **120** can be continued, when the fine water is discharged and stopped, the controller **150** controls the switch **143** of the electrifying portion **140** so that the switch **143** is closed or opened, and switches the fine water generating unit **130** between the electrified state and the non-electrified state to discharge and stop the fine water.

Further, in the above-described embodiment and each of the above-described modification examples, the fine water discharge device **100** discharges the fine water formed of the water cluster by the fine water generating unit **130** (fine water discharging element **10**) into the second space **S2** as it is. In addition to this, for example, a device for converting the water cluster into plasma, for example, corona discharge, creeping discharge, or the like, which is well-known low-temperature plasma generation means, is provided, and thus, active oxygen ($\text{OH}^\cdot$ radical) having strong oxidizing power can also be generated efficiently from water cluster (fine water). As described above, the active oxygen ($\text{OH}^\cdot$ radical) generated from the water cluster (fine water) has strong oxidizing power. For example, when the fine water discharge device **100** (fine water discharging element **10**) is applied to an air purifier, it can be effective in sterilizing bacteria, preventing insects, and preventing mold.

Further, in the above-described embodiment and each of the above-described modification examples, the power supply **142** of the electrifying portion **140** is assumed to be a commercial power supply. Therefore, the fine water dis-

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charge device **100** of the above-described embodiment and each of the above-described modification examples is a stationary type device. Instead of this, by using a battery capable of charging and discharging as the power supply **142**, the fine water discharge device **100** can be made portable (so-called cordless type device).

Further, in the above-described embodiment and each of the above-described modification examples, the user directly operates the controller **150** of the fine water discharge device **100**. Instead of or in addition to this, for example, the user can operate a remote controller which enables a remote control of the controller **150**, or set a timer for an operation time and a stop time of the controller **150** so as to operate the fine water discharge device **100**. Alternatively, when the fine water discharge device **100** includes a motion sensor, a temperature/humidity sensor, or the like, the controller **150** can operate the fine water discharge device **100** according to detection results of these sensors.

Further, in the above-described embodiment and each of the above-described modification examples, the fine water discharge device **100** is applied to the beauty equipment so that the fine water can effectively infiltrate the skin of the user. As described above, the fine water generated and discharged by the fine water discharge device **100** (fine water discharging element **10**) is the bound water which has the particle size of 50 nanometers or less and is uncharged (non-electrified). Therefore, the fine water discharge device **100** can be applied to other than the beauty equipment.

For example, when the fine water discharge device **100** (fine water discharging element **10**) is applied to a dryer to discharge the fine water to a scalp or hair of the user, the fine water can infiltrate the scalp or hair and apply a moisturizing effect to the scalp or hair. As a result, for example, improvement of hair gloss, prevention of itchiness, prevention of dandruff, prevention of hair loss, or the like can be expected. Further, when the fine water discharge device **100** (fine water discharging element **10**) is applied to a hair growth promoter, infiltration of a hair growth agent applied to the scalp of the user can be promoted, and the promotion of hair growth can be expected.

Further, when the fine water discharge device **100** (fine water discharging element **10**) is applied to a drug delivery device to discharge the fine water to a skin to which the drug is applied, a percutaneous absorption of the drug can be promoted, and for example, improvement in effects of a drug with respect to various diseases can be expected. Further, when the fine water discharge device **100** (fine water discharging element **10**) is applied to a dry eye prevention device (eye mask or the like) to discharge the fine water to eyes or a face of the user, it can be expected that the infiltration of fine water suppresses dry of the eyes. Further, when the fine water discharge device **100** (fine water discharging element **10**) is applied to an inhaler or a respirator to discharge the fine water to a mucous membrane or the like in an oral cavity of the user, it is expected that a moisturizing effect and a humidifying effect will be improved by the infiltration of the fine water into the mucous membrane or the like.

Further, in these cases, the fine water generated by the fine water discharge device **100** (fine water discharging element **10**) is directly discharged to the user (human). Instead of this, it is also possible to discharge the fine water generated by the fine water discharge device **100** (fine water discharging element **10**) to a non-human object. For example, when the fine water discharge device **100** (fine water discharging element **10**) is applied to a refrigerator or a pantry to discharge the fine water into food, since the fine water is

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uncharged (non-electrified) bound water, a moisturizing effect of stored food can be enhanced. As a result, for example, it can be expected to maintain a freshness of the food refrigerated in the refrigerator and improve flavor of the food stored in the pantry.

Further, when the fine water discharge device **100** (fine water discharging element **10**) is applied to a plant growing device to discharge the fine water to a plant, for example, it can be expected to promote a growth of the seedling by discharging the fine water to the seedling. Further, when the fine water discharge device **100** (fine water discharging element **10**) is applied to a static eliminator to discharge the fine water to an industrial product or the like, since the fine water is uncharged (non-electrified), for example, it can be expected to have an effect of removing static electricity from paper, a semiconductor, a paint, or the like.

The invention claimed is:

1. A fine water discharging element capable of transitioning between an adsorption state where water contained in a fluid to be treated is adsorbed and a discharge state where the adsorbed water is discharged to the fluid to be treated, the fine water discharging element comprising:

- a base material portion;
- a plurality of particles having a core-shell structure including a core portion forming a nucleus and a shell portion formed of a polymer material having a polar functional group capable of hydrogen bonding to cover the core portion; and
- a nanochannel smaller in width than said particles, is formed between the shell portions of the plurality of particles by laminating the plurality of particles on an outer surface of the base material portion in a film shape, wherein

the fine water discharging element is transitioned between the adsorption state and the discharged state by changing the temperature of the water in the nanochannel by controlling an electrifying of at least one of the base material portion and the plurality of particles.

2. The fine water discharging element as set forth in claim 1, wherein the shell portion includes, as the polar functional group, at least one of a sulfonic acid group ($-\text{SO}_3\text{H}$), a carboxyl group ($-\text{COOH}$), a hydroxyl group ($-\text{OH}$), an amino group ($-\text{NH}_2$), an amid group ($-\text{C}=\text{ONH}-$), and a pyrrolidone group (NCOC_3H_6).

3. The fine water discharging element as set forth in claim 1, wherein the shell portion includes, as the polymer material, at least one of poly styrene sulfonic acid, polyvinyl sulfonic acid, polyvinyl pyrrolidone, polyvinyl alcohol, and sodium polyacrylate.

4. The fine water discharging element as set forth in claim 1, wherein the core portion is formed of polymer material or inorganic material.

5. The fine water discharging element as set forth in claim 1, wherein the base material portion is formed by at least one of stainless-steel based metal, a copper-based metal, a carbon material, a conductive ceramic material and a conductive resin material.

6. The fine water discharging element as set forth in claim 1, wherein the base material portion is a metal foil formed of stainless-steel based metal, and the particles are PEDOT/PSS constituted by the core portion formed of poly (3,4-ethylenedioxythiophene) and the shell portion formed of poly (styrene sulfonic acid).

7. The fine water discharging element as set forth in claim 6, wherein the PEDOT/PSS is set so that a weight ratio between ethylenedioxythiophene (EDOT), which is a monomer of the PEDOT, and the PSS is between 1:10 and 1:3.

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8. The fine water discharging element as set forth in claim 1, wherein a thickness of the plurality of particles by laminating the plurality of particles on the outer surface of the base material portion in the film shape is 1 μm ~30 μm .

9. A fine water discharge device comprising:

a fine water generating unit comprising the fine water discharging element as set forth in claim 1;

a flow path arranged in the fine water generating unit and allowing a first space and a second space to communicate with each other;

an air blowing unit introducing air in the first space into the flow path and discharging the air introduced into the flow path into the second space;

an electrifying portion electrically connected to the base material portion of the fine water generating unit to electrify the base material portion; and

a control portion which controls the air blowing unit and the electrifying portion, wherein the control portion includes:

a discharge control portion which executes a discharge mode of introducing the air in the first space into the flow path by the air blowing unit, electrifying the base

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material portion of the fine water generating unit by the electrifying portion to discharge the water adsorbed on the particles of the fine water generating unit to the air introduced into the flow path and discharging the fine water to the second space together with the air introduced into the flow path by the air blowing unit; and a discharge preparation control portion which executes a discharge preparation mode of introducing the air in the first space into the flow path by the air blowing unit and non- electrifying the base material portion by the electrifying portion to adsorb the water introduced into the flow path to the particles.

10. The fine water discharge device as set forth in claim 9, wherein the base material portion of the fine water discharging element is formed to have a honeycomb shape in structure.

11. The fine water discharging element as set forth in claim 9, wherein in the discharge mode, a temperature of the electrified base material portion increases by a range of 20° C. to 50° C. as compared with that of the non-electrified base material portion.

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